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OntoBPR: An ontology-based framework for performing building permit reviews using standardized information containers

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ABSTRACT

Building permitting is essential for ensuring the safety, sustainability, and societal alignment of construction projects. Despite interest from both practitioners and researchers, the process remains largely manual and fragmented. Ontologies offer a promising solution by managing complexity and enabling automation through semantic information, though current ontologies in the building permit domain are limited to specific aspects like building code checking. On the process level, the OntoBPR framework integrates multiple domain-specific ontologies for a seamless digital permitting process and provides a workflow to automate the lifecycle of the permit review. Therefore, it suggests integrating the submitted building application using standardized information containers. The paper explores how digital applications can be submitted, reviewed, verified for completeness, and forwarded to authorities, and how permit review results can be gathered to support decision-making and automate notification issuance, and it provides a demonstration in a case study. In conclusion, OntoBPR formalizes a multi-layered ontology that advances and aligns the partitioned building permit process and provides an adaptable framework to harmonize diverse legal, informatics, and procedural aspects.

1. Introduction

Building permitting is a critical component of any construction project [1]. It serves as a gatekeeper, ensuring that the built environment is safe, sustainable, and aligned with societal needs [2]. The digitalization of the building permit process holds the promise of addressing the shortage of skilled workers and simplifying the management of increasingly complex processes and regulations [3].

Despite growing interest from both practice and research, the process remains largely manual or minimally digitalized, and although much effort has been put into the development of technical support of different steps within the building permit review, the entire process remains fragmented. Most research concentrates on Automated Compliance Checking (ACC), but building permitting is to be seen as a much larger process that involves many different stakeholders, heterogeneous data, and complex regulations. Although solutions for ACC have been proposed by many researchers over the past decades [4], those checking mechanisms need to be aligned with the broader picture. Otherwise, they would fail their purpose within the scope of building permitting.

A review by Bloch and Fauth [5] highlights that while many technical solutions are discussed in studies, they seldom align with the

actual processes involved in building permit reviews. Ontologies offer a promising approach to managing complexity and enabling automated processing by providing semantic information that can harmonize processes and other types of information [6]. Ontologies enable the exchange of information between stakeholders within the building permit review process while still maintaining the flexibility to adapt the data scheme also for further permit systems in other countries. Ontologies are published as Findable, Accessible, Interoperable & Reusable (FAIR) data and are usually available as open-source artifacts, allowing all stakeholders in the process, in particular the applicants, to maintain a highly transparent procedure. Ontology alignment is crucial because it enables different systems to “speak the same language”, ensuring consistency in how concepts and relationships are defined and understood across various domains. For instance, in automated regulatory compliance checking, aligning Building Information Models (BIM) with regulatory ontologies ensures that both use the same concepts to describe the same objects, thereby facilitating accurate and efficient compliance verification [7,8]. Such alignments not only reduce ambiguity but also enhance the ability to automatically check

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for regulatory compliance by harmonizing the interpretations used by different systems.

However, in the context of building permitting, only a few ontologies exist (e.g., [9–11]), each covering different aspects of the overall process, and most of them relate to the building code checking. This results in a still fragmented process that lacks comprehensive integration. Therefore, the modeling and execution of standardized processes of the entire building permit processes coupled with the information needed in each process step delivered by the applicant is required. Comprehensive integration for documents and data can be achieved by employing the Information Container for linked Document Delivery (ICDD) [12]. The ICDD is a container that provides a semantic layer for document delivery by using Resource Description Format (RDF) and internationally standardized ontologies describing the container structure and the metadata of the content [13]. ICDD is standardized in ISO 21597 [13] and is proven to be validated and checked against different kinds of rules using the Shapes Constraint Language (SHACL) [14] so that the use for the building application submission and building permit review is possible.

Our approach is reusing different ontologies to cover the entire building permit process in building authorities. The reuse is conceptualized in horizontal integration and fusion of multiple *domain-specific* ontologies, which describe relevant information and knowledge in the domain of building permitting. The resulting ontological framework called *Ontology-based Building Permit Review (OntoBPR)* integrates heterogeneous datasets and enables interoperability among systems and automated reasoning to support decision-making across the building permit review process – from submitting a building application until the issuance of a notification letter – orchestrated by building permit authorities.

In summary, the research contribution of OntoBPR lies in its development of a formalized, multi-layered ontology that advances the theoretical foundations of digital building permitting. By integrating semantic technologies such as ICDD containers for standardized data exchange and SHACL rules for automated compliance checks, OntoBPR bridges gaps between disparate regulatory, technical, and administrative domains. Furthermore, its adaptable upper ontology provides a novel approach to harmonizing diverse legal and procedural frameworks, contributing to knowledge in ontology engineering and decision support systems in building permitting.

This article is structured into a background section in Section 2 for highlighting previous efforts on semantic descriptions of building permits and the use of information containers, a methodology section in Section 3 for providing background about the materials and methods used for this research as well as a requirements elicitation. Moreover, the paper contains a section on preliminary work on OntoBPR in Section 4 and a concept section in Section 5 for introducing the concept of this research based on the building permit review process. It continues with an implementation section in Section 6, an evaluation section in Section 7 proving the framework with a demonstration using an exemplary building application, and eventually providing a critical discussion with implications and limitations in Section 8 and the conclusions in Section 9.

2. Background and literature review

The building permit process is a critical regulatory mechanism designed to ensure that construction projects comply with local building codes, zoning laws, and safety standards [15]. The digitalization of the building permit process represents a transformative shift towards greater efficiency, transparency, and accessibility in construction project management [16,17]. In a comparative study of the building permit processes in Europe, the most common upper-level processes were defined as formal review, assignment, participation of other agencies, content check, and issuing notification letter [18]. By leveraging advanced technologies such as BIM, Geographic Information

Systems (GIS), and cloud-based platforms, among others, municipalities can automate and streamline various stages of the building permit process [19]. These digital tools facilitate the electronic submission, review, and approval of building permit applications, significantly reducing paperwork and processing times. Moreover, digital platforms enhance communication and collaboration among stakeholders, including architects, engineers, contractors, and regulatory authorities, by providing real-time access to project data and status updates [20]. The implementation of algorithms further optimizes the building permit review by automatically checking compliance with building codes and zoning regulations, e.g., [21,22], and [23]. However, despite these advancements, the process remains fragmented due to the diverse regulatory environments and varying levels of technological adoption across different jurisdictions. Inconsistent standards, limited interoperability between systems, and resistance to change from traditional methods contribute to the uneven implementation of digital solutions [24]. Moreover, there are country-specific open formats like XBau in Germany for providing structured information about the building permit, which, however, are not integrated with BIM and information containers and thus again display only a decoupled solution [25]. This fragmentation can hinder the full realization of digitalization benefits, underscoring the need for more standardized approaches and collaborative efforts to align processes across jurisdictions at the national, regional, or municipal levels.

2.1. Semantic description of knowledge and concepts for building permitting

Ontologies create a semantic model of a specified domain, representing a segment of the real world [26,27]. Studer et al. [28] define an ontology as “a formal, explicit specification of a shared conceptualization”. These specified models are described using semantic web languages, primarily the RDF, which structures data in triples of subject, predicate, and object, forming RDF graphs [29]. RDF is augmented by more expressive languages like the RDF Schema (RDF-S) and the Web Ontology Language (OWL). RDF-S adds to RDF by defining classes and properties, while OWL allows for setting constraints among these elements [30]. The semantic web utilizes SPARQL Protocol and RDF Query Language (SPARQL) for querying and manipulating asserted facts adhering to ontological terms (so-called Assertion-Box (A-Box) data [29]), and rule languages like the Semantic Web Rule Language (SWRL), SPARQL Inference Notation (SPIN), and the Shapes Constraint Language (SHACL) for rule implementation [31,32]. For BIM data in the form of the standardized Industry Foundation Classes (IFC) models, a transformation of IFC to a semantically rich representation in OWL, the so-called ifcOWL, has been proposed [33–35] and converters are publicly available [36]. Further ontologies that provide a modular approach for representing, e.g., the topology of a building as the Building Topology Ontology (BOT) [37] or properties of a building or structure as the Ontologies for Property Management (OPM) [38], are developed and maintained in the context of the World Wide Web Consortium (W3C) Linked Building Data (LBD) Community Group and form a foundation for the usage of ontologies for modeling semantic building data in the web.

2.2. Ontologies in building permitting

Ontologies in building permitting formalize knowledge-intensive processes and regulations, enhancing efficiency and interoperability by providing a structured framework for data exchange [39]. They define relevant concepts and relationships, enabling seamless communication among systems and stakeholders. In permit reviews, ontologies standardize compliance criteria, improving processing speed, reducing ambiguities, and supporting decision-making. Several ontologies for building permits have been developed and are reviewed below.

Fauth et al. [40] created a generic taxonomy classifying terms of building permit systems. The taxonomy provides a shared concept

Table 1
Overview of available ontologies for the building permit review.

Author	Ref.	Country	Domain	Users/ Use cases	Limitations
Fauth et al. (2024)	[40]	International	Overall building permit system description (buildings)	Overall users in building permit process	Only taxonomy, does not conceptualize an ontology
Nuyts et al. (2024)	[10]	Belgium (Flanders)	Code compliance checking with rules (buildings)	Code compliance checking	Limited to Flemish accessibility rules
Zhong et al. (2018)	[44]	China	Building regulations for environmental monitoring	Code compliance checking	Limited to rules on environmental monitoring in China
Jiang et al. (2022)	[8]	International	Code checking and completeness checks	Code compliance checking and completeness check	Limited to specific regulatory requirements, based on IFC
Grisiute et al. (2023)	[41]	Singapore	Allowable Gross Floor Area (GFA) for plots	Code compliance checking	Limited to Gross Floor Area rules in Singapore
Silvennoinen et al. (2023)	[42]	Singapore	Zoning checks (buildings)	Code compliance checking	Limited to zoning checks in Singapore
AEC3PO	[9]	Europe	Compliance checking	Building compliance, used in real-world scenarios	Requires alignment with standards, potential issues identified during evaluation
Fitkau et al. (2024)	[11]	Germany	Fire safety (buildings)	Code compliance checking	Limited to fire safety regulations based on German model building
Ma et al. (2024)	[45]	China	BIM-QC Ontology	Code compliance checking with building code	Discussion: Limited expressivity of the rule language

of building permit systems, including a hierarchical structure for the components and definitions for each class. So far, this research is limited to concepts and hierarchical organization and does not include relations between concepts and formalization. Therefore, terms and hierarchical concepts should be reused in the approach of the research at hand.

Dirdi et al. [9] present AEC3PO, an ontology developed within the ACCORD project to represent building regulations and permitting processes. Composed of modular components, AEC3PO aligns with standards like IFC, DCMI Metadata Terms, and the European Legislation Identifier (ELI) to support automated compliance checks and interoperability.

Grisiute et al. [41] developed a semantic spatial policy model to estimate allowable Gross Floor Area (GFA) for plots in Singapore. Using ontologies and Knowledge Graphs (KG), they standardized regulatory concepts and stored them in a geospatial KG. This eliminates manual calculations and enhances regulatory data interoperability.

Silvennoinen et al. [42] present OntoZoning, an ontology for land use regulations in Singapore's urban planning. It provides a structured, machine-readable format for zoning regulations and land use concepts, improving data clarity and accessibility. OntoZoning supports urban data integration and analytics, enhancing efficiency and interoperability in planning processes.

Nuyts et al. investigate the use of SHACL for automated compliance checking of building models. Their research addresses inefficiencies in traditional methods by making legislative requirements machine-readable, thereby enhancing automation and accuracy. The approach is adaptable to various prescriptive building legislations, ensuring comprehensive data validation and quality assurance within building projects [10,43]. Fitkau and Hartmann [11] developed the Fire Safety Ontology (FiSa) to integrate fire safety into digital building design. FiSa formalizes fire safety knowledge and German regulations (MBO), enabling structured compliance checks and early detection of design issues. It incorporates preventive and structural measures, using rules to infer building classes and conditions, highlighting the potential of ontology-based systems to enhance fire safety compliance.

Zhong et al. [44] introduced the Building Regulation Ontology within a framework for environmental monitoring and compliance

checking in BIM. This framework integrates building information, sensor data, and regulatory information through four ontologies: Building Information Ontology, SSN Ontology, Building Regulation Ontology, and Building Environmental Monitoring Ontology. While effective in an energy station project, it focuses on environmental aspects and lacks a comprehensive representation of broader building regulations.

The code ontology introduced by Jiang et al. [8] is central to an automated code compliance-checking methodology, comprising four ontologies: code, design model, merged, and compliance-checking ontologies. It captures knowledge from building codes but focuses mainly on information organization, limiting its adaptability to diverse regulatory systems and automation for AEC projects.

Another ontology for compliance checking is provided by Ma et al. [45]. They developed the BIM model Quality Checking ontology (BIM-QC), which applies building code rules using SWRL based on OWL instances derived from IFC models. While BIM-QC can process rules from building codes like those in China, a major limitation is that not all rules can be compiled into SWRL. This reflects a broader challenge in formalizing building regulations, but the specific reliance of SWRL on OWL-DL and its limited support for procedural or geometric reasoning makes it less suitable for certain rule types commonly found in construction regulations. Thus, a rule language can only be fully utilized in combined approaches with geometric processing engines such as presented in Pauwels et al. [46].

This section presents an overview of existing ontologies and their limitations as reported in Table 1, highlighting the need for a new, more comprehensive framework. Despite substantial efforts in developing ontologies for the building permit process, current models only capture specific parts of the procedure. A comprehensive, cross-process ontology is still missing. The focus of existing ontologies is primarily on task-specific aspects, such as building reviews and compliance checks, and they are often tailored to specific countries rather than being generic. A significant advancement would be to align these ontologies for broader applications. The current state of building permit ontologies is fragmented and limited in scope. Developing a new, comprehensive ontology that bridges these gaps will significantly enhance the efficiency and effectiveness of the building permit process. This study sets the groundwork for such an ontology, emphasizing the need for alignment and integration of existing models for greater utility.

2.3. Container-based information exchange for building permitting

The need for information containers in building permits has already been identified by Grünz et al. [47] in 2001. They compared the permit process to a regular business process and stated that the highly distributed stakeholders, the legally binding documentation, and the heterogeneity of different platforms, data formats, and data processing techniques necessitate information containers [47].

Information containers in digital construction projects are standardized in ISO 21597, detailing the Information Container for linked Document Delivery (ICDD) [13]. The ICDD standard aims to establish a vendor-neutral data structure for exchanging diverse building data in the construction industry. This standard defines a container structure that accommodates the exchange payload and facilitates distributed data integration using Linked Data (LD) and Semantic Web [13]. The LD approach enables seamless connectivity to web standards like HTTP and REST. Semantic Web technologies, such as SPARQL queries, offer efficient data retrieval capabilities within the context of the ICDD container [14]. ICDD enables modeling arbitrary data using an ontological layer stored in the container or as web resources [48]. First approaches have identified the ICDD as a data exchange mechanism usable for building permits [49]. For the structural assessment during the building permit review, a use case has been defined for employing ICDD and openBIM formats for seismic assessment and structural e-permits [50, 51].

2.4. Research gaps and development of research questions

So far, existing ontologies have primarily focused on isolated aspects of the building permit process, such as the representation and checking of fire safety regulations, zoning laws, or accessibility requirements within specific national contexts (see Table 1). Although these domain-specific ontologies provide valuable contributions, they do not adequately capture the building permit process holistically and across multiple regulatory domains. Furthermore, the existing approaches from the literature are typically developed for specific legal systems, making them difficult to transfer to other countries or to generalize beyond their original use case because it is not evident to what extent these ontologies are transferable to other contexts.

Based on the literature reviewed in this section, several gaps in the current state of the art have been identified, which are reported in the following:

- country-specific solutions to data exchange for building permits lack applicability in other countries and are not compatible with each other
- the usage of integrated data exchange formats for building application documents (e.g., for submitting interconnected IFC, PDF, XML documents in a context annotated package) is limited
- processes for building permit review in countries are analyzed but the subprocesses are not aligned, thus a generalized solution for implementing processes is required
- existing ontologies have only focused on specific aspects within their respective building permit processes
- building codes and standards are not interconnected with the processes and stakeholders

As a result, there is no comprehensive, ontology-based representation of the building permit process that integrates regulatory requirements, process steps, data exchange formats, and the involvement of different stakeholders. To overcome these limitations, OntoBPR aims to develop a generalized, ontology-based representation of the building permit process that is not confined to a single regulation set or national context. Instead, it will be designed to flexibly represent different regulatory frameworks and process models in a unified structure, enabling both national and international applicability. By integrating regulatory

requirements, process steps, involved stakeholders, and data exchange formats into a coherent ontological framework, OntoBPR intends to create a foundation for the digitization, automation, and standardization of building permit procedures. Building on these identified gaps, this research explores the following key questions, which address the submission, verification, and processing of digital building applications, by focusing on the role of ontologies and information containers in enabling automation and international applicability:

- RQ1 *How can building permit processes, building code compliance checking, and the building permit application be integrated conceptually into a unified framework?*
- RQ2 *How can the completeness and regulatory compliance of digital building applications be automatically verified through an ontology-based framework that links process steps, stakeholders, and regulatory requirements?*
- RQ3 *How can the ontology-based framework support the automated processing of building permit reviews, including decision-making documentation and the automated generation of official notification letters?*

Addressing these research questions will contribute to the development of a streamlined, efficient, and transparent digital building permit process. By leveraging ontological frameworks, standardized information containers, automating verification using SHACL validation and forwarding processes using SHACL reasoning, and integrating decision support systems with automated notifications, the entire life cycle of building permits can be optimized, ultimately fostering a more effective regulatory environment.

3. Methodology

The methodology for developing OntoBPR in this research relies on the Linked Open Terms (LOT) methodology proposed by Poveda-Villalón et al. [52] and is adapted as depicted in Fig. 1. The LOT methodology originally encompasses four distinct stages illustrated: ontology requirement specification, ontology implementation, ontology publication, and ontology maintenance that are tailored for industrial ontology implementations. In this research we broke the methodology down into three steps, that are extracted from the LOT flow and supplemented by a flow of artifacts between these steps.

First, the requirements engineering is conducted based on the Ontology Requirement Specification Document (ORSDD) [53], which includes defining the use case specification, identifying the purpose and scope, and specifying both functional and non-functional requirements. The formalization of the ORSD encapsulates and documents the determined requirements for the ontology, as detailed in Section 5.2. In the ORSD, previous works on the OntoBPR framework are considered, and requirements for the required vocabulary are elicited after the processes are analyzed. Requirements are elicited as competency questions (CQs) as proposed for ontology development by Ren et al. [54] and evaluation by Bezerra et al. [55].

After specifying the requirements for the OntoBPR framework in Section 5, the methodology proceeds to the second step of the framework implementation, as outlined in Section 6. This phase involves the conceptualization of the ontology, which includes creating classes, class hierarchies, relationships, attributes, property hierarchies, and axioms. The potential for reusing existing ontologies is also considered according to the best practices defined in [56,57]. After the conceptualization, the ontology is encoded using OWL and RDFS and documented. This step involves publishing the ontology. Once documented using Hypertext Markup Language (HTML) and the PyLODE documenting tool [58], it is stored in an online repository [59] on GitHub. GitHub also serves as a platform for ongoing ontology maintenance, where bugs can be detected, and the ontology can be updated with new requirements.

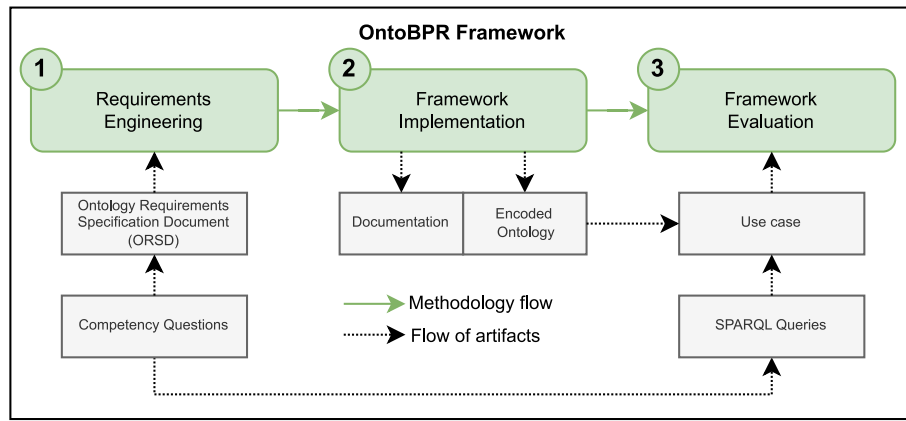


Fig. 1. Methodology developed for the implementation of the OntoBPR framework.

Eventually, the OntoBPR framework is evaluated in this paper according to [60] through consistency checking, criteria-based evaluation, and process-based evaluation. Therefore, a use case is defined for demonstrating the overall framework and evaluating it based on the developed competency questions from the requirements engineering. Since the overall framework also consists of the integration of Information Container for linked Document Delivery (ICDD) containers for delivery and forwarding the building application throughout the permit review process, the previous work of Hagedorn [12] is used, in which a web platform is developed that is capable of maintaining ICDD containers as part of a Common Data Environment (CDE) and enables features of the Semantic Web, such as querying using SPARQL or validating using SHACL [14]. The platform is utilized for the OntoBPR approach and will be used for the evaluation of the overall process.

In this methodology, we emphasize the importance of ontology reuse as a key factor for achieving efficient and interoperable knowledge representation within the building permit review. Encouraging reuse not only helps minimize unnecessary duplication but also supports a collaborative approach to continuously enhancing semantic resources, contributing to the long-term sustainability of ontology-based systems [61]. Therefore, our research considers the principles of the Onto4Reuse framework to facilitate this reuse process [62]. In addition to purely reusing existing ontologies, the development of upper ontologies plays a crucial role in this research for providing a common conceptual process that enables seamless integration of multiple modular domain ontologies [63]. Upper ontologies define fundamental, high-level concepts that are independent of specific domains, offering a standardized semantic framework that promotes consistency across different knowledge domains [64]. By establishing a shared upper ontology for the overall building permit review process, modular domain ontologies for representing knowledge about dedicated legislation or checking aspects can be more effectively aligned, ensuring interoperability while still allowing each module to evolve independently to meet specific requirements.

4. Preliminary work on the OntoBPR framework

In this section, an overview of preliminary work on the OntoBPR framework, the OBPA ontology, and the IDDO ontology is given. The previous study by the authors [39] presented the ontology-based building permit review (OntoBPR) as a workflow to avert digital disruptions in the building permit review process by mapping the process ontologically. The OntoBPR framework is introduced, aligning two existing ontologies and validated through a case study with SHACL rules.

Within the conducted alignment, new properties were introduced, establishing the connection between the OBPA and the IDDO. This connection can be used to show which guidelines and standards need to be checked in the building application and by which building authority.

The workflow checks building code requirements comprehensively, providing immediate feedback during permit submission for legal certainty, defect prevention, and error reduction. The paper emphasizes the need for ontologies in formal review and suggests expansions for numerical data testing [65]. Besides the extensions of the overall process to further process steps, the OBPA and IDDO ontology are refined for this work, as described in the following sections.

4.1. Ontology for building permit authorities (OBPA)

The OBPA handles the knowledge and information of building authority, including building officials and the building application [66]. The ontology is designed as a domain ontology and can, therefore, be specified by any building authority for the application. Furthermore, the ontology is related to upper-level ontologies like the Friend of a Friend Ontology (FOAF) [67] or vCard [68] to ensure clarity of terminology, understanding, and reusability of the ontology. OBPA is based on three main entities *obpa:BuildingAuthority*, *obpa:Person*, and *obpa:BuildingApplication*.

The process of assigning a *obpa:BuildingApplication* to a worker is based on the worker roles, skills, and responsibilities represented as *obpa:Person* and hired by the *obpa:BuildingAuthority*. To describe the task of a person assigned to a building application, activities are linked to both the person and the building permit using the *obpa:Activity* class. The activities can be further specified by subclasses, to categorize and specify the activities. By developing constraints, the approval process can be ensured to follow a specific sequence of steps, enabling compliance with established test sequences and avoiding procedural errors.

The ontology can be adapted to specific building authorities, enabling a flexible and expandable system to represent the heterogeneous process of obtaining building permits and the assignment processes and structures within building authorities. Furthermore, it is possible to add additional components or to remove components that are not needed.

4.2. Properties and information requirements

The initial development of IDDO was aimed at converting the knowledge embedded in building codes and construction guidelines into a digital format, structuring it within a hierarchical tree [65,69]. The organization of this ontology aligns with the data model and management rules set out in ISO 23386 [70], encompassing both general metadata and specific management details associated with so called groups of properties and the properties. Each property is further described by attributes that define its values, including the scope of values, units, options from predefined lists and arrays, connections to other properties, and the capability for dynamic value calculation [65].

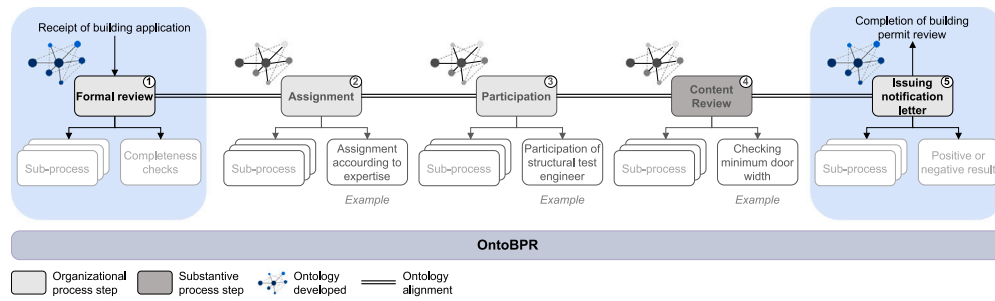


Fig. 2. Building permit process and mapping to ontological data schemas.

In the research work by Mellenthin Filardo et al. [69], IDDO was renamed to the ISOProps Ontology and harmonized with the Level of Information Need (LOIN) Ontology by Liu et al. [71], as well as the Data Template Ontology (DT) developed in the cited work, to facilitate a standardized approach to specifying construction information. This harmonization enables the specification of information needs at the Building Element level. Following the alignment, the ISOProps ontology underwent structural changes, particularly in handling property boundary values, aligning with ISO 23387 principles. Additionally, the management of dimensions, physical quantities, and units has been streamlined over the DT and the IDDO to centralize the information on physical quantities. The subsequent assignment of properties collected in the data template to the corresponding Feature of Interest (FOI), such as *dice:Equipment* or *bot:Building*, is still realized through the OPM property state pattern [38], as previously described by Zentgraf et al. [65].

5. Concept and requirements specification

The development of a framework and its implementation for a holistic use of ontologies within the building permit process, as shown in Fig. 2, is the focus of this study. Therefore, the developments based on the building permit process are mainly based on the findings of previous research by Fauth [72]. The holistic nature of the approach compared to the preliminary work is achieved by the fact that all process steps are mapped ontologically, that ICDD containers are used to exchange the building application information, that the check in the (1) *Formal Review* step is supplemented by using SHACL and the results are provided in the (5) *Issuing Notification Letter* step.

The preliminary work is described first before the ontologies are supplemented and the ICDD containers are integrated. From Fig. 2, the building permit process can be seen at a low level of detail, while the detailed process can be found in Fauth [72]. The process starts with receiving the building application, which will be exchanged as an ICDD container. Next, the (1) *Formal Review* process will be conducted, which is one of the main parts of this research. The reviewed container is processed to the (2) *Assignment* step, where the container is forwarded to the respective responsible building official followed by the (3) *Participation* step. (2) *Assignment* and (3) *Participation* are already covered by the previous works on the Ontology for Building Permit Authorities (OBPA) (see Section 4.1). These processes are followed by the (4) *Content Review* for which the authors used the ISOProps ontology (see Section 4.2) to represent properties in buildings and validate models. Based on the results of (2), (3), and (4), the last step of the process is the (5) *Issuing notification letter*, which has not been part of previous work and is extended into the OntoBPR framework in this research. After that, the building permit review is completed.

In order to describe the entire building permit process ontologically, the ontologies are linked horizontally as presented in Fig. 2. Each process step is mapped and represented by an individual domain ontology.

In the context of the OntoBPR, these ontologies can be seen as modules, which are horizontally segmented and aligned to each other. Section 6 provides a detailed description of how the individual process steps are connected on an ontological level. Each horizontal module adds domain-specific terms, information, and knowledge to the OntoBPR framework. Therefore, a new horizontal module extends the concept of the OntoBPR by adding new classes and properties [73]. The horizontal alignment ensures the consistency, coherence, independence, and separation of the modules [74,75] by providing the relationships of the modules in separated alignments [76].

5.1. Process-based concept

The building permit review main process steps are depicted in Fig. 3 in a Business Process Modeling and Notation (BPMN) diagram. As already stated, OntoBPR has, before its extension, not yet covered the full process. Therefore, the OBPA will be extended to represent the missing process steps. The diagram shows that the process always starts with the (1) *Formal Review* for checking whether the required information is provided and administrative compliance is met. Next, (2) *Assignment* of a suitable building official, (3) *Participation* of other involved agencies, and (4) *Content Review* checking whether the substantive compliance is met can happen either in parallel or after one another. In any case, process steps (1), (2), (3), and (4) must be finished before the process (5) *Issuing Notification Letter* can start.

After the building permit application is received, the (1) *Formal Review* is undertaken. Since OntoBPR fosters a data-driven approach rather than a document-based approach, the application is represented as a set of information structured in an ICDD container.

5.1.1. Step (1) Formal Review

The (1) *Formal Review* covers the two subprocesses (1.1) *Completeness Check* and (1.2) *Review Preparation*, as illustrated in Fig. 4.

In the (1.1) *Completeness Check*, the ICDD is proved against its completeness (i.e., all required information is available), and if the information quality for the administrative information is assured (i.e., is the correct information in the correct place in the ICDD structure). Therefore, an additional metadata schema extending the already existing ICDD metadata for the documents in the container is developed in this research as part of OntoBPR, which can be employed by validation rules that entail the information about required documents. The metadata can be extended and adjusted for different authorities receiving the building applications. In case the application passes the (1.1) *Completeness Check* positively, the (1.2) *Review Preparation* starts. If the *Completeness Check* is not successful, a follow-up request for the required documents or information can be initiated directly. The (1.2) *Review Preparation* enables processing steps (2), (3), and (4) and builds the connection to them based on the information given by the building permit application.

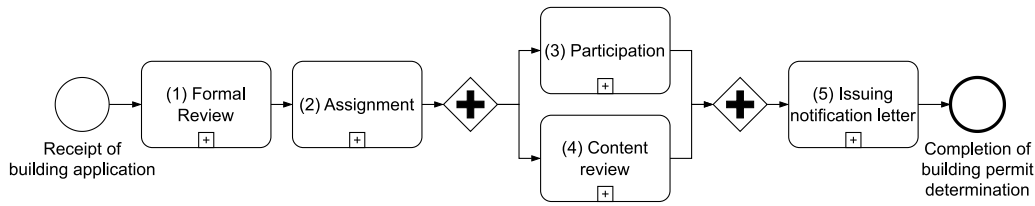


Fig. 3. Building permit review main processes.

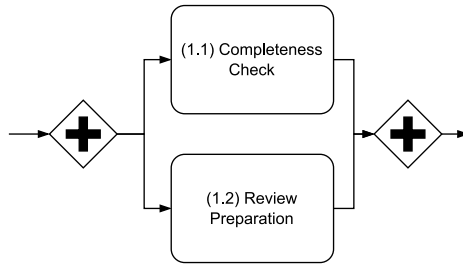


Fig. 4. Formal Review subprocesses.

5.1.2. Steps (1–3) Assignment, Participation, Content Review

For preparing (2) *Assignment*, interconnection to the assignment process is built in terms of outlaying the personnel and organizational information from the individual authority. In preparation for (3) *Participation*, relevant information from the ICDD will be related to concerns of other agencies of public interest. This might refer to geospatial information as an external reference (e.g., the distance to a nature preservation area concluding an involvement of the respective nature preservation authority as described in [20]). For the preparation of (4) *Content Review*, OntoBPR runs a check which regulations across different codes and ordinances need to be contently checked based on specific information and indicators provided with the building permit application (see Fig. 5, Layer 1–4). That refers to the fact that different kinds of buildings require different checks based on different codes and ordinances (e.g., high-rise buildings).

Preparing (4) *Content Review*, OntoBPR proceeds with a selection of the relevant regulations within step (1.2) in the form of a multiple-layer structure as shown in Fig. 5. Based on the indicators and information (e.g., type of building, location of the building), specific regulations are pointed out to be checked by applying ISOProps building code dictionaries. The layers range from superficial to detailed. For example, depending on the state where the building is located, a federal or regional building code might be considered; depending on the city, municipal statutes might apply, or the local development plan regulates the area where the building is to be built.

Other indicators such as a gathering area within a building (which can apply to different types of buildings), a specific ordinance might become relevant for the content check (e.g., in Germany, the “Versammlungsstättenverordnung” [Ordinance on Places of Assembly] for rooms with a specific number of square meters and with a capacity of a specific number of persons). Multiple countries also use a classification system for buildings, which leads to the necessity of checking specific regulations or specifying the responsibility of applying the required regulations (which might not be within the respective municipality).

Fig. 5 represents a schematic multi-layer structure for a permit review process up to compliance checking. The figure indicates that all subsequent layers contribute to or support this review process. While Layers 1–4 are necessary to identify Applicable Requirements, Layers 5–6 perform the actual compliance check. Layer 1, titled “Legislations”, includes fundamental legal frameworks - Planning Law and Building Law. Layer 2, labeled “Regulatory Documents”, comprises relevant

documents such as the Development Plan, National Planning Policy, Building Code, Building Ordinances, and Building By-laws. In the third layer, various thematic areas of regulation are highlighted, such as Zoning, Land Use, Fire Safety, and High-rise regulations, among others. These themes represent different aspects of planning law and building law that are subject to regulation. The fourth layer focuses on specific clauses within these themes, such as Set Back Line, Escape Route, and Door Width. These clauses represent detailed requirements or stipulations that must be adhered to within the broader regulatory themes. The fifth layer addresses Information Requirements, where key aspects like Existence, Datatype, and Unit are considered. These factors are checked for compliance against the previously identified clauses and themes. Finally, the sixth layer involves detailed Rule Checking, which includes categories such as Mathematical, Relational, Conditional, and Value Range checks. On the right side of the diagram, the process flow is indicated through arrows, outlining that Layer 1 is required for (2) *Assignment*, Layer 1–3 for (3) *Participation*, and Layer 1–6 for (4) *Content Review*.

Once the substeps (1.1) and (1.2) are completed, the steps (2), (3), and (4) will be performed as described in OntoBPR [39]. The steps are executed on the foundation of the submitted ICDD container, including the relevant building application information. All steps are documented within the ICDD. Detailed implementation of these steps employing the information containers are provided in Section 6.

5.1.3. Step (5) Issuing Notification Letter

As soon as the previous process steps have been completed, there are three options for how the process (5) *Issuing notification letter* can be conducted, which are depicted in Fig. 6.

The result of the permit process can be evaluated as (a) *positive*, (b) *positive under conditions*, and (c) *negative (with justification)*. These results are accumulated over the results of the SHACL validations performed on the data of the building application in the previous steps. All reports are materialized in the ICDD container. Particularly, the appearance of SHACL violations is considered a negative outcome of the application. Therefore, the SHACL severities *sh:Info*, *sh:Warning*, and *sh:Violations* need to be considered already carefully when applying the validation shapes to the data. The resulting validation report will be stored in the ICDD container as justification for a negative outcome and returned to the applicant afterward. In case of a positive outcome under conditions, the validation reports are provided in the container as well. Additional information and justifications of the participating actors in the review process can be stored in the container during the process. Manually reported results and documentation can be recorded in the container as well to maintain high transparency of the overall process.

5.2. Requirements specification for the OntoBPR extension

For the requirements specification, the Ontology Requirements Specification Document (ORS) is utilized as introduced in the methodology (see Section 3). In the specification phase, the primary objective is to provide the aim and scope of the OntoBPR. This involves identifying the potential users and outlining the ontology’s specific functional and non-functional requirements. The specifications of the OntoBPR are

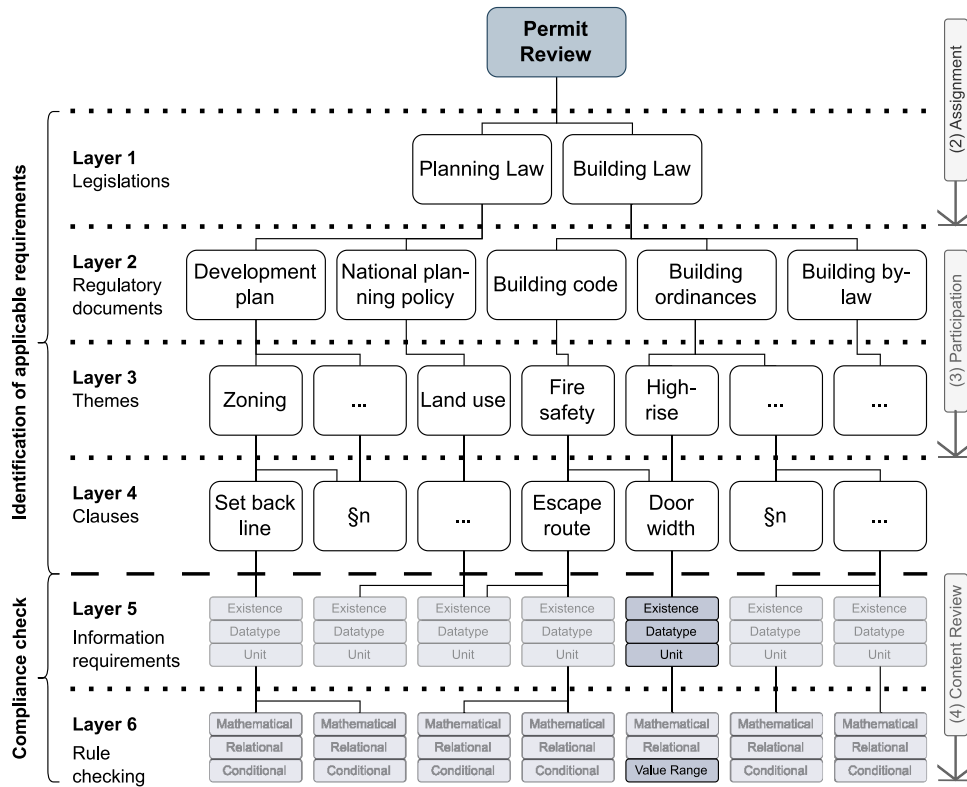


Fig. 5. Schematic multi-layer structure for a permit review process.

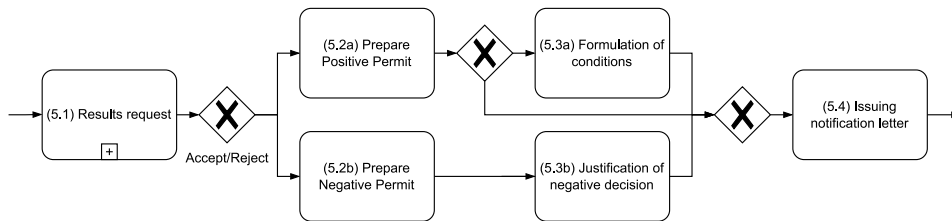


Fig. 6. Sub-processes of the main process (5) Issuing notification letter.

achieved by methodically answering a series of pertinent questions that guide the specification process. The formalized competency questions (CQs) are summarized in the ORSD, reported in Table 12 in Appendix.

The ORSD covers the purpose scope, users, use cases, and non-functional requirements for the overall building permit review process. The functional requirements are only taken into account for the two missing steps: the formal review and the issuing of the notification letter.

The ontology should form the structural basis for supporting the building permit review process by using and adapting existing ontologies to ensure a transparent, efficient, and objective administrative process that complies with legal requirements. It provides a detailed data schema representing all relevant data and information required during the building permit review process. OntoBPR aims to provide a fundamental basis for systems that can reduce the workload of building officials through semi-automated checks, thereby speeding up and improving the inspection process. The standardization and clear data structuring increase the traceability and transparency of decisions and reduce the risk of errors. In addition, OntoBPR enables better collaboration and communication between the various parties involved in the building permit review process, as all parties have access to a standardized knowledge base that contains all relevant information needed for each of the five building permit review process steps.

During the development of the ontology, three groups of end-users and five intended use cases emerged. The first group of end-users involved in all five use cases are the building authorities, who maintain and process the building permit review. Within the participation process (UC 3), the building authorities integrate additional specialized inspectors as a third end-user group. The latter end-users, considered more passive, are the applicants who have submitted the building application. Use Case 5 (issue of notification letter) is particularly relevant for these end-users, as it provides information on the outcome of the building permit review. The intended uses identified during the specification phase are the five process steps of the Building Permit Review. These five use cases are further detailed in Section 5.1.

The next step in the specification process was to define the non-technical requirements for the OntoBPR. These requirements ensure that the ontology is not only technically correct but also practically applicable and usable. They cover several aspects that help to ensure the ontology's quality, efficiency, and usability. Coverage and Sufficiency refer to the breadth and depth of the domain representation of the ontology. They ensure that all relevant aspects of the application domain are covered comprehensively and in detail. This is important to ensure that the ontology represents all the necessary information and relationships within the application domain. Another important requirement is consistency, which ensures that the ontology

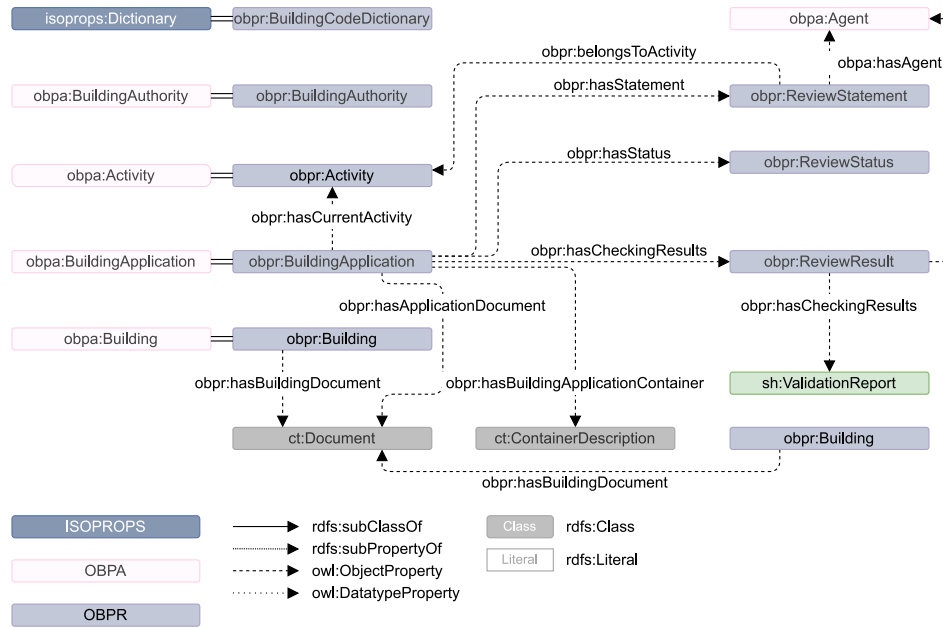


Fig. 7. Major extensions to OntoBPR and class equivalencies to OBPA and ISOProps.

remains free of contradictions and conflicts. All definitions and relationships within the ontology must be consistent and logically coherent to provide reliable and correct results. The clarity and conciseness underline the ontology's comprehensibility and brevity of presentation. Information must be presented clearly and concisely to avoid misunderstandings and enable users to understand it quickly. Reusability and extensibility refer to the ability of the ontology to be customized for different purposes and its capacity for extension. The need for transparency emphasizes the importance of the ontology being comprehensible and transparent in its structure and logic. The functionality and underlying principles of the ontology must be easy to understand to promote trust and acceptance among users. Ontology modularity focuses on the design of the ontology in discrete, manageable sections. This modularity facilitates understanding and maintenance by dividing the ontology into manageable modules, thus simplifying handling and further development.

The final step in the specification of the ontology requirements was to define the functional requirements arising from the Formal Review and the Issuing Notification Letter for the ontology (see Table 12 in the Appendix). These requirements are formulated in the form of competence questions, which are explained below. The following competence questions assess the completeness and suitability of the submitted Formal Review. The evaluation starts by determining whether the building application has been delivered in an ICDD container (CQ1.1). Following this, it is verified if the required information to process the application is provided within the ICDD (CQ1.2).

Subsequently, the next questions involve checking if the information available is sufficient to initiate the subsequent procedural steps, namely Step 2, Step 3, and Step 4 of the building permit review (CQ1.4). This encompasses, for example, the information needed to assign the project to one or more building officials.

The second group of competence questions evaluates the availability and clarity of results of steps 2 to four to create a proper Issuing Notification Letter. Therefore, it is necessary to check if the intermediate results from Step 2, Step 3, and Step 4 are available (CQ5.1). Following this, the focus shifts to determining whether a clear final result can be derived from these interim results, categorizing the outcome as either positive, positive with conditions, or negative with justification (CQ5.2).

In conclusion, the functional requirements clearly indicate that the planned extension of the ontology to include (1) *Formal review* and (5)

Issuing notification letter is heavily influenced by the steps already modeled in the ontology, namely Steps 2 to 4. Therefore, it is crucial to pay special attention to these dependencies during the extension process to ensure that the result remains a homogeneous and continuous process, thereby achieving the desired efficiency improvements in the Building Permit Review process.

6. OntoBPR implementation

The overall architecture and implementation of the extended and aligned OntoBPR framework is given with the updated ontology and extensions provided in the already established ontologies (see Section 6.1). It is introduced how the building permit process is supported by ICDD containers and how these containers can be implemented for submitting and reviewing a digital building permit application. Moreover, the different logics and functionalities of the participation procedure and the determination of responsible persons are described by employing the OBPA, while the actual checking process is facilitated by using the digitalized building codes in the ISOProps scheme. Eventually, this section presents how the OntoBPR framework can be implemented into a web-based platform that allows for information exchange based on the ICDD standard (see Section 6.3).

6.1. Framework extension

Extensions in the OBPA are provided in this research based on the conceptual developments from the previous chapter. Extended vocabulary that leads to the ontologies no longer being able to be used independently of each other is created in the OntoBPR alignment in the *obpr*: namespace that is declared in the preliminary work by the authors [39]. An overview of extensions is available in Fig. 7 and will be described in detail in the further course of the section.

Overall, multiple terms are aligned or reused with the OntoBPR. First of all, the class equivalence between *isoprops:Dictionary* and *obpr:BuildingCodeDictionary* is established. Further equivalences are established between *obpa:BuildingAuthority* and *obpr:BuildingAuthority*, between *obpa:Activity* and *obpr:Activity*, as well as between *obpa:BuildingApplication* and *obpr:BuildingApplication*. Particularly, the class equivalence between the two activity classes is helpful for the definition of sub-activities in the further course in the OntoBPR as provided in Fig. 8.

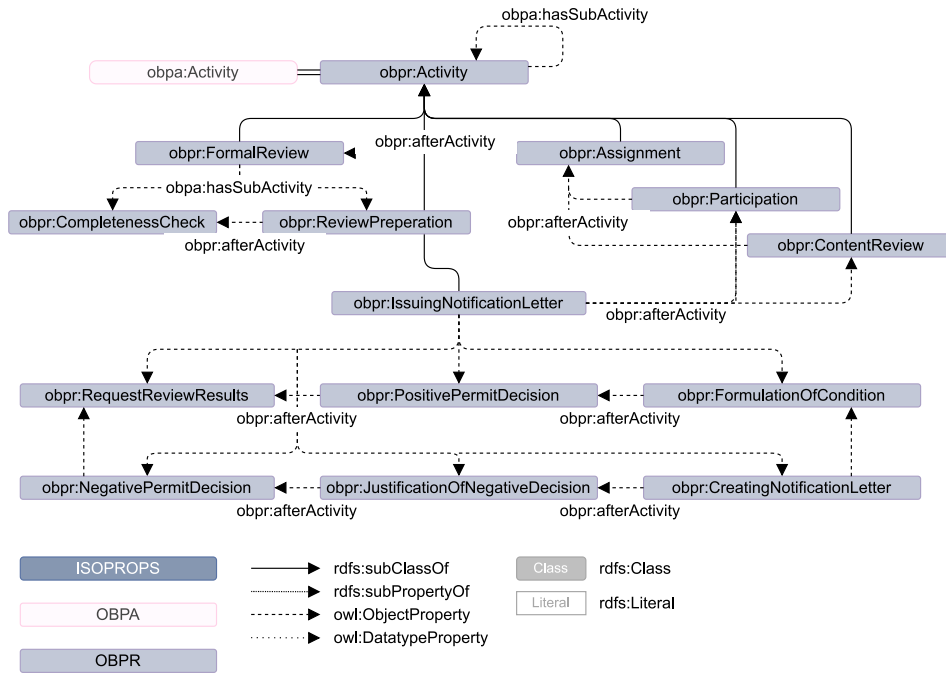


Fig. 8. Implemented Process-Driven Building Permit Review in OntoBPR.

Based on the concept and objectives of this paper, the most important class expressions (1) *Formal review* and (5) *Issuing notification letter* are identified and added to the OntoBPR ontology as *obpr:FormalReview* and *obpr:IssuingNotificationLetter* (see Fig. 8). Both classes are subclasses of the *obpr:Activity* class.

6.1.1. (1) Formal Review

The *obpr:FormalReview* class is integrated into the OntoBPR ontology with an additional object property *obpr:review* with domain *obpr:FormalReview* and range *obpa:BuildingApplication* to add semantic rich meaning to the relation between the review and the application. The *obpr:FormalReview* class is a subclass of *obpa:Activity*. In the next step, the existing class *obpa:BuildingApplication* is related to a building application container via the object property *obpr:hasBuildingApplicationContainer* to an instance of *ct:ContainerDescription*. The *ct:ContainerDescription* class semantically represents the index of an ICDD container. For submitting building applications, ICDD containers are instantiated also on a file-based level, including the semantic descriptions of the building application using the OntoBPR framework.

Before the container can be processed in the formal review process step towards the (1.1) *Completeness Check* and the (1.2) *Review Preparation* as provided in Fig. 4, interrelations to the transported payload documents in the ICDD container must be specified. The first mapping is defined between the *obpr:Building* and an instance of *ct:Document* as *obpr:hasBuildingDocument* (see Fig. 7). Another interrelation is provided between the *obpr:BuildingApplication* and an instance of *ct:Document* in the ICDD container as *obpr:hasBuildingApplicationDocument*. Further mappings can be applied using a general object property *obpr:hasPayloadDocument* but vary between types of application and can be adapted as sub-properties for further required documents. The object *obpr:hasPayloadDocument* is also the super-property of *obpr:hasBuildingApplicationDocument* and *obpr:hasBuildingDocument*.

The activities (1.1) *Completeness Check* and (1.2) *Review Preparation* are modeled as subclasses of *obpa:Activity* providing the class *obpr:CompletenessCheck* and *obpr:ReviewPreparation*. Instances of these two classes can be related to the instances of the super-process *obpr:FormalReview* by using the *obpa:hasSubActivity* object property inherited from the equivalent superclass *obpa:Activity*. Moreover, a property *obpr:afterActivity* is introduced to establish a procedural context for

activities. Restrictions have been formulated in order to maintain the predefined process structure as provided in Section 5. After defining the activities relevant to the formal review, requirements must also be specified for the ICDD container that is submitted as a replacement for the paper or digital application.

The ICDD container that should be submitted by the applicant must contain the elements reported in Table 2, in which the required documents result from the building code of the state of North-Rhine Westphalia, Germany from BauO NRW 2018 [77]. The table contains the typical ICDD data structure (index, ontology resources, payload documents, and payload triples) and provides the information of the required documents at each permit stage and for the submission as a matrix. The presence of a document or dataset in the container is marked with a checkmark, while the document must not be used for each step that is marked. Furthermore, the implementation of the ICDD that is used and described in Section 6.3 does not allow the deletion of contents permanently for the sake of seamless documentation. Deviations from this container profile depicted in Table 2 can be modeled depending on the state or country, while completeness of the documents also varies depending on the type of building or type of application. Thus, the table only depicts an exemplary ICDD container, while the vocabulary to be used within the container is fixed by the applied ontologies.

For simplicity, the container can already be defined as a template for the applicant so that it directly contains all relevant ontologies. In the template, the index of the application is already instantiated in the *BuildingApplicationIndex.ttl* RDF graph, containing the process steps and relevant structure as well as the main *obpr:BuildingApplication* entity, which can be filled with the respective information of the application. Since applications are still submitted in PDF form nowadays, the structure provided with the ontologies in the application index can be used to create a machine-readable application in the future. All necessary certificates, the site plan, the respective building model and the PDF of the building application must be integrated into the container before the submission of the container. These payload documents are then linked within the container in the linkset *Doc_Application_Links.rdf* and additional relations are established between the *obpr:BuildingApplication* and the *ct:Document* entities, e.g., using the *obpr:hasBuildingDocument* or the *obpr:hasBuildingApplicationDocument* object properties.

Table 2
ICDD container contents throughout the different stages of building permit review.

Information container				Permit stages				
Location	File	Type	Subm.	1	2	3	4	5
./								
	Index	.rdf	✓	✓	✓	✓	✓	✓
./Ontology Resources								
	Container	.rdf	✓	✓	✓	✓	✓	✓
	Linkset	.rdf	✓	✓	✓	✓	✓	✓
	ExtendedLinkset	.rdf	✓	✓	✓	✓	✓	✓
	OBPA	.rdf	✓	✓	✓	✓	✓	✓
	ISOProps	.rdf	✓	✓	✓	✓	✓	✓
	OntoBPR	.rdf	✓	✓	✓	✓	✓	✓
./Payload documents								
	BuildingApplication	.pdf	✓	✓	✓	✓	✓	✓
	BuildingApplicationIndex	.ttl	✓	✓	✓	✓	✓	✓
	BuildingModel	.ifc	✓	✓	✓	✓	✓	✓
	SitePlan	.pdf	✓	✓	✓	✓	✓	✓
/certificates/	SoundInsulation	.pdf	✓	✓	✓	✓	✓	✓
/certificates/	ThermalInsulation	.pdf	✓	✓	✓	✓	✓	✓
/certificates/	Stability	.pdf	✓	✓	✓	✓	✓	✓
/certificates/	Accessibility	.pdf	✓	✓	✓	✓	✓	✓
	BuildingApplicationLog	.ttl		✓	✓	✓	✓	✓
	BuildingAuthorityKG	.ttl			✓	✓	✓	✓
/compliance/	Assignments	.ttl			✓	✓	✓	✓
/compliance/	NatureConservancy.shapes	.ttl				✓	✓	✓
/compliance/	NatureConservancy.report	.ttl				✓	✓	✓
/compliance/	Accessibility.shapes	.ttl					✓	✓
/compliance/	Accessibility.report	.ttl					✓	✓
/issuance/	Decision	.ttl						✓
/issuance/	NotificationLetter	.pdf						✓
./Payload triples								
	Doc_Application_Links	.rdf	✓	✓	✓	✓	✓	✓

In step (1) *Formal Review*, the file *BuildingApplicationLog.ttl* is added to the container. This file contains an RDF graph of *obpr:Review-Statement* instances that are referenced by the main index entity *obpr:BuildingApplication* of the building application in the *BuildingApplication-Index.ttl* graph via the *obpr:hasStatement* object property. These statements are used to track the changes and events occurring during the permit review inside the ICDD container to maintain a fully transparent process. An exemplary statement could be logged for the successful submission of the application container or for the request of missing information in the (1) *Formal Review* step. This application log can be used for further analysis of the building permit process, e.g., to perform process analysis or process optimization tasks. Each statement is also referable to the respective person who authored the statement, is related to the process step, and has a time stamp attached.

Moreover, the *obpr:BuildingApplication* entity is equipped with a status and a current activity, while each instance of the activities class also has a flag completed that indicates a boolean value whether the activity is completed and subsequent activities can be started. The status of the building application is based on the overall process execution and drawn decisions and can take the values *obpr:Accepted*, *obpr:Accepted-WithConditions*, *obpr:Rejected*, *obpr:RequestInformation*, and *obpr:Under-Review*. From the submission, the *obpr:UnderReview* status is assigned until an event occurs that changes the status. Each status change is additionally logged as a statement inside the *BuildingApplicationLog.ttl*.

6.1.2. (2) Assignment

Based on the content review, the building officials will process the building application in step (2) *Assignment*. To enable a transparent and objective review, the assignment is done using rules. In addition, rule-based assignment ensures consistent building reviewer utilization

and that the reviewer's skill is sufficient to handle the building permit. Therefore, the respective instances of the OBPA representing the responsible building authority are loaded into the container based on the information provided in the application and stored in the *BuildingAuthorityKG.ttl*. The respective instances in the *BuildingAuthorityKG.ttl* are retrieved from an external graph database and integrated into the container for further processing. Moreover, it is also possible, to maintain the graph in a remote location and reference to that location from the container to always keep the information up-to-date.

The *BuildingAuthorityKG.ttl* comprises several key components and modules that facilitate the representation and management of various entities and their interrelationships of the authority based on the OBPA ontology, which is aligned with the OntoBPR. As presented in previous research [66], the core entities include *obpa:BuildingAuthority*, which represents the overarching authority responsible for building permits; *obpa:Person*, representing individuals involved in the process such as building officials, applicants, and other relevant personnel. Via the class equivalence that is introduced between *obpa:BuildingApplication* and *obpr:BuildingApplication* in this paper, the interconnection between the application in the *BuildingApplicationIndex.ttl* to the authorities is provided. The rule module of the OBPA comprises rules for decision-making processes, which can be general or specific to a particular authority.

These rules are defined using SHACL and facilitate the automated or assisted assignment of building applications to the appropriate building officials. The definition of rules as SHACL shapes using RDF enables us to manage different rule sets of authorities as named graphs, for instance, in remote locations. Hagedorn [12] has shown how SHACL rules can be shared between different authorities as web resources to be reused in different projects or adapted to specific requirements

without being bound to a specific implementation, since every SHACL processor can be utilized to execute them on any dataset. SHACL shapes for the assignment and participation of the building application are difficult to automatically generate and have to be adapted for their use with specific authorities. This necessitates expertise in Semantic Web and SHACL but has to be done only for an initial setup, for which an overview of shapes can be found in previous research by Fauth et al. [66]. So that these rules can be applied to the container, the building application index must contain relevant information on the building type, e.g., special building, and the location or address of the building, e.g., the zip code, for determining the responsible authority. If available, further application properties may also be utilized for the assignment.

After executing the rules, the resulting inferred assignments are instantiated within the *Assignments.ttl* RDF graph and establish the interconnection between the application and the responsible agents in the building authorities. In this process step, the files *Assignments.ttl* and *BuildingAuthorityKG.ttl* are incorporated into the ICDD container to supplement the existing RDF graphs.

6.1.3. (3) Participation

The step (3) *Participation* in building permit reviews involves identifying and assigning relevant Authorities of Public Interest (AoPI) to a building application. This step ensures that necessary AoPIs, such as specialized municipal authorities for the natural conservancy, are engaged when required. Fauth [72] highlights inefficiencies, where too many or too few AoPIs are involved, causing delays and inflated procedures due to additional document reviews. To address these issues, SHACL rules and Decision Modeling Notation (DMN) [78] tables are used to semi-automate the assignment of AoPIs as provided in the research by Fauth et al. [79]. This helps streamline the decision-making process by linking business modeling with decision logic. Additionally, the OBPA framework has been expanded in the original work [79] to include new roles representing various AoPIs, such as the *obpa:PublicOfficer* or the *obpa:ProofingStructuralEngineer*. Organizations involved in the process are modeled as subclasses of *obpa:Organization*. This structured approach aims to improve efficiency and reduce delays in the building permit review process.

As a result of the (3) *Participation*, the assigned AoPIs are added to the *Assignments.ttl* RDF graph, and for each assigned organization, the respective code dictionary, including SHACL shapes for validation, is incorporated into the container, which in this example in Table 2, is referred to as the *NatureConservancy.shapes.ttl* for checking natural preserving rules and after the checking, a respective SHACL report file is incorporated into the container for each performed domain-specific check, e.g., the *NatureConservancy.report.ttl*. An example of a checking report will be provided in the case study in Section 7.

6.1.4. (4) Content Review

According to Fig. 5 in the conceptualization, the (4) *Content Review* requires the assignment and participation to be decided. Thus, it requires decisions about the application of legislation, regulatory documents, and applicable themes and clauses in these documents. Based on these, requirements and rules can be formulated. Therefore, the utilized building codes are represented by the ISOProps ontology as part of an *obpr:BuildingCodeDictionary*, which is an equivalent class to *isoprops:Dictionary* and through the specification of the ISOProps dictionary, it is a subclass of an *dc:Catalog*, which is defined in the Data Catalog Vocabulary (DCAT). The instances of *obpr:BuildingCodeDictionary* are attached to the *obpr:BuildingApplication* using the *obpr:applies* object property that is already introduced in previous work [39].

Furthermore, an interconnection between the building code dictionary and the building authority is established using the *obpr:used_by* object property. As presented in previous works, the properties in the building code dictionary can be automatically translated into SHACL shapes [39]. The instances of the building codes can be external sources

that are linked to the ICDD container of the application as an external resource. The derived SHACL shapes are incorporated into the ICDD container, e.g., as the *Accessibility.shapes.ttl* for the accessibility code rules (see Table 2).

The SHACL shape instances are aggregated into a *obpr:ShaclShapesSet* that has the connection to the instances of *sh:Shape* via the *obpr:hasShapes* object property (see Fig. 9). The *obpr:ShaclShapesSet* is connected to the respective *obpr:BuildingCodeDictionary* via the object property *obpr:hasShapesSet*. The SHACL shapes can be executed on the aggregated dataset in the ICDD container and equivalent to the checks performed by an external participating institution from the step (3) *Participation*, the results of the SHACL validation are recorded into the container as *Accessibility.report.ttl*. Depending on the building codes and regulations, multiple building code dictionaries and their corresponding shape sets can be incorporated into the container. The validation reports are linked to the *obpr:BuildingApplication* via the object property *obpr:hasCheckingResults* and an instance of the class *obpr:ReviewResult* that has a reference to the agent, that is responsible for conducting the required check. The review results reported using SHACL can be supplemented by manual entries extending the SHACL report, so also checks not performed using SHACL have results reported into the ICDD container.

We acknowledge the fact that SHACL validation shapes are not available out of the box for building codes and usually experts are considered for developing these shapes, but also want to reference further publications that automatically extract SHACL shapes from regulations using natural language processing, such as Donkers and Petrova [80] or Zentgraf et al. [65], or enrich smart standard definitions with validation rules already in the standardization bodies, such as Zentgraf et al. [81].

6.1.5. (5) Issuing Notification Letter

Eventually, step (5) *Issuing Notification Letter* concludes the building permit review. Therefore, using SPARQL, all validation reports that are incorporated into the container during the (3) *Participation* and (4) *Content Review* steps are queried in the first sub-process step. Since step (5) *Issuing Notification Letter* implements two exclusive gateways in the process diagram, based on the results queried from the validation report, the first decision, whether a positive permit can be issued, is made. This decision right now is made by an actor on behalf of the building authority. An automatic decision is not possible at this stage of implementation because also the review results can be weighted differently regarding their severity.

If the decision is to reject the application, the status of the application is set to rejected, and the violations from the reports and the statements from the building application log are retrieved to justify the negative decision. Justification statements referred to as *obpr:Justification* are a subtype of the *obpr:ReviewStatement* class and are attached to the rejected *obpr:BuildingApplication* via the *obpr:hasStatement* object property.

On the other path, when the building application is accepted, the status will either be set to accepted or to accepted under conditions. The subsequent decision, whether conditions are made, is again drawn by the responsible person in the building authority. If it is decided to add conditional acceptance, then in another process step, the conditions are defined before the final issuance of a notification letter. These processes are also defined as subclasses of the general *obpr:Activity* class and as sub-activities of the *obpr:IssuingNotificationLetter* class, as it is depicted in Fig. 8. Condition statements referred to as *obpr:Condition* are a subtype of the *obpr:ReviewStatement* class and are attached to the accepted *obpr:BuildingApplication* via the *obpr:hasStatement* object property.

In either case, the decisions as *obpr:Justification* or as *obpr:Condition* instances are integrated into the ICDD container into the *Decisions.ttl* RDF graph that is stored in the *issuance* folder. From the decisions RDF graph in conjunction with the other RDF graphs in the container, a notification letter is generated and stored in the *NotificationLetter.pdf* file inside the container. The notification letter is generated based on

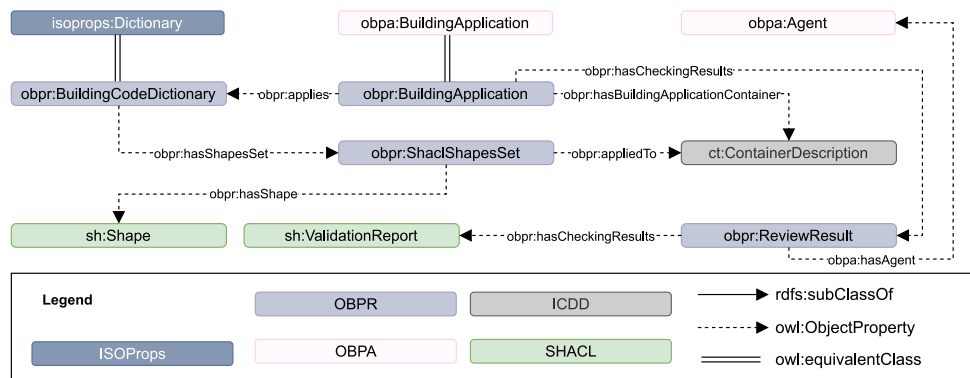


Fig. 9. OntoBPR content review extension classes and interrelations to ontologies.

a hard-coded HTML template that is filled with data from the RDF graphs retrieved via a SPARQL query and then exported as a PDF or alternatively can directly be stored into machine-readable formats using RDF, JSON, or XML. Although more reproducible methods, such as the SPARQL Template Transformation Language (STTL) [82,83] exist, in this research, the hard-coding of the template was chosen, due to the lack of implementation of STTL into SPARQL engines running together with the RUB ICDD platform. However, this step of generating templates from the data in the ICDD containers can be followed up in future work.

After the step (5) *Issuing Notification Letter* is finished, the container and all its contents can be returned to the applicants. Due to the consequent track of changes and events in the building application log, the applicants have a transparent decision at their disposal.

6.2. Evaluation of OntoBPR

This section describes the evaluation of OntoBPR, detailing the methods and results to ensure that both functional requirements (FRs) and non-functional requirements (NFRs) are satisfied [84].

A consistency check of OntoBPR was conducted by using the Pellet reasoner [85] and the Ontology Pitfall Scanner (OOPS) [86]. The Pellet reasoner is an open-source OWL-DL reasoner used to ensure the consistency of an ontology and to infer new knowledge when applied to instance data. The OOPS tool, instead, is used to automatically detect conceptualization errors in ontologies based on the defined OOPS principles [86] including, for instance, inconsistencies or redundantly defined classes, missing hierarchical relationships, and inconsistent property expressions [86]. An overview of all possible pitfalls checked by the OOPS is provided by Poveda-Villalón et al. [86].

During the evaluation of OntoBPR with OOPS, after two iterations, all relevant pitfalls were eliminated. In 14 cases, minor issues were indicated for missing explicitly declared inverse relationships (defined as pitfall P13 in [86]). For the brevity of the ontology, these inverse relationships were not included in the OntoBPR ontology. The remaining pitfalls on missing annotations (defined as pitfall P08 in [86]) and untyped classes (defined as pitfall P34 in [86]) concern imported vocabulary, e.g., from schema.org or the ICDD ontology, and are therefore not further considered for refining the OntoBPR. Eventually, based on both tools, the OntoBPR is positively validated against consistency.

Eventually, reusability is provided by the OntoBPR ontology as one of the FAIR principles for data sharing. Therefore, the ontology is completely documented and annotated with English metadata according to best practices. All ontology entities are defined in a permanent namespace for the ontology. Moreover, the ontology and its documentation are hosted on GitHub,¹ where collaborators can propose extensions

or additions using forks and pull requests and raise issues about the ontology in the ticketing system. Thus, reusability is preserved, and the ontology can be developed and improved continuously.

6.3. Implementation into a web-platform prototype

The web-platform prototype utilized in this research was originally developed in the work by Hagedorn [12,87] and is extended with the OntoBPR framework during this research. The RUB ICDD Platform facilitates data sharing, exchange, and visualization among stakeholders via ICDD information containers according to ISO 21597 [13]. It supports the management of container statuses in compliance with ISO 19650-1 [88] and provides tools for data querying, manipulation, and integration from relational databases. Data is stored in file-based containers and managed using an open-source ICDD framework [89]. The extensible platform supports multiple client interfaces through its REST API, allowing seamless collaboration and data exchange across various applications [90]. The REST API is publicly documented as the OpenICDD interface [91].

For analysis and validation of ICDD containers, the platform also implements SPARQL querying and SHACL validation using the dotnetRDF library [92]. To query and validate multiple RDF graphs, dotnetRDF enables aggregate graphs in an in-memory triple store and executes SPARQL queries or validates SHACL shapes while RDFS reasoning is implemented into the ICDD platform.

7. Case study

The following section presents a use case to demonstrate the feasibility and usability of the developed architecture and the implementation of the extended and aligned OntoBPR framework. The focus of the case study is on the two use cases defined in the ORSD in Table 12, namely for the steps (1) *Formal Review* as UC1 and (5) *Issuing Notification Letter* as UC5. Regarding the ontology development procedure according to the LOT methodology, these two use cases will additionally be evaluated with competency questions, while the existing process steps in between will be demonstrated to provide the entire process in this case study. This case study utilizes a fictitious building permit authority and construction project in Germany.

In the following example, the OntoBPR framework is applied to a case study of a public building in Germany. The case focuses on the review process for a public office building (see Fig. 10) on the university campus in Bochum, a city in North Rhine-Westphalia (NRW) in Germany.

The building has 2500 m² and is located in a proximity of 450 m to water bodies (i.e. a lake in this case study). Public buildings are characterized as special constructions that require the involvement of specific external parties and compliance with specific regulations. The case study illustrates the practical implementation of the framework as previously described in Section 6 in the ICDD platform and

¹ <https://rub-informatik-im-bauwesen.github.io/ontobpr/>, last accessed: 12.03.2025.

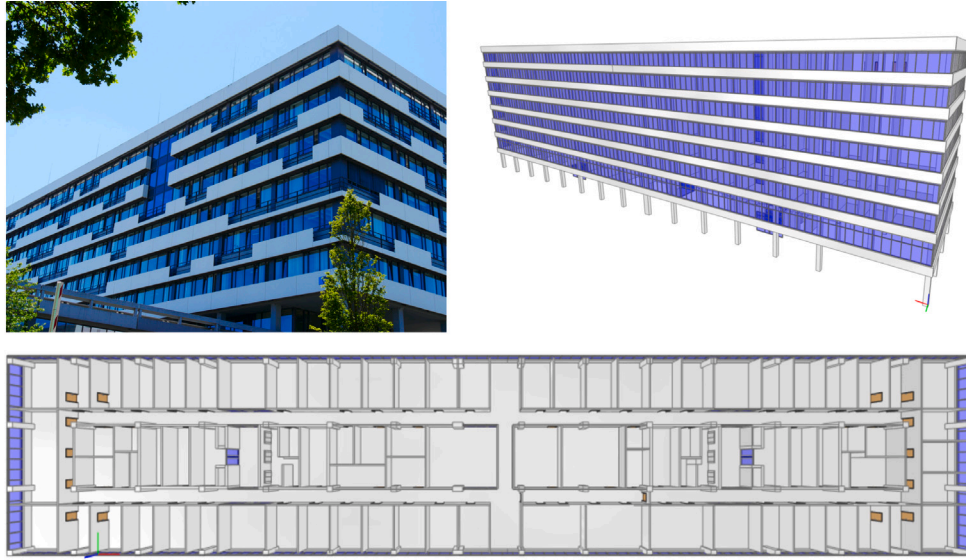


Fig. 10. Case study building at the university campus in Bochum.

Table 3
Ontologies and namespaces used in the case study.

Prefix	URI
:	http://example.org/ontobpr-usecase#
ct:	https://standards.iso.org/iso/21597/-1/ed-1/en/Container#
iProp :	http://example.com/iProp#
obpa:	https://w3id.org/obpa#
isoprops:	https://w3id.org/isoprops#
ontobpr:	https://w3id.org/ontobpr#
owl:	http://www.w3.org/2002/07/owl#
rdf:	http://www.w3.org/1999/02/22-rdf-syntax-ns#
rdfs:	http://www.w3.org/2000/01/rdf-schema#
vcard:	http://www.w3.org/2006/vcard/ns#
xsd:	http://www.w3.org/2001/XMLSchema#
sh:	http://www.w3.org/ns/shacl#
props:	https://w3id.org/props#

demonstrates the implemented extensions. It utilizes several different ontologies that are listed in Table 3 and are used in the SPARQL queries for the ontology evaluation. Therefore, the definition of prefixes in the SPARQL queries is omitted.

7.1. (1) Formal Review

The case study involves an analysis of the information provided via the ICDD in step (1) *Formal Review*, which includes a completeness check according to the requirements for building applications as defined in the national BauPrüfVO 1995 [93] and the review preparation (see again Fig. 4). Key elements identified at this stage include the location of the building (provided through the address of the building provided in the building application) and its classification as a public building (provided in the building application).

As described in the corresponding Section 5.1, the (1) *Formal Review* consists of two independent sub-steps, the Completeness Check and the Review Preparation. The building application is provided in the ICDD container, submitted via the implemented ICDD platform, as depicted in Fig. 11. On the left-hand side of the Figure, the full structure of the building application container is displayed. The container is structured according to the structure proposed in Table 2 as defined in the previous section. The ICDD container includes the relevant information, and the successful application submission is already logged into the building application log (center). The building model (right) is visualized from the submitted data.

The main input to the (1) *Formal Review* process is the building application handed over to the building authority by the applicant. To guarantee the usability of the developed OntoBPR framework for the formal review and its sub-processes, three competency questions were formulated in advance, listed in Table 4, and will be evaluated on the dataset provided in the ICDD container in the following. If the competency questions can be answered with the help of the developed framework and its vocabulary, it can be concluded that the framework is evaluated to fulfill UC1.

The evaluation of the Formal Review begins by determining whether the applicant submitted the building application in an ICDD container (CQ1.1). To be able to verify this, it must be checked whether a *ct:ContainerDescription*, which semantically represents the index of the ICDD container, is assigned to the building application represented by an instance of the class *obpa:BuildingApplication*. The relation between these two instances is mapped via the object property *obpr:hasBuildingApplicationContainer*. Listing 1 and 2 in the Appendix display a SPARQL query and a SHACL rule that can be utilized to answer this question (CQ1.1). The displayed SPARQL query retrieves information about a building application and its associated containers. Specifically, it searches for all instances of building applications and their corresponding containers, where each *obpr:BuildingApplication* instance must have exactly one *ct:ContainerDescription* instance linked to it. The query result is a list of building applications, each paired with exactly one associated container, ensuring that only those applications with a single, uniquely linked container are returned. In case of an empty result set, the corresponding building application does not contain a valid ICDD container.

The SHACL rule illustrated herein follows an equivalent approach, defining constraints for the class *obpr:BuildingApplication*. It is stated that each object of the class *obpr:BuildingApplication* must be linked to a building application container. This relationship is defined by the object property *obpr:hasBuildingApplicationContainer*, with a minimum and maximum of one link permitted. Additionally, the container in question must belong to the class *ct:ContainerDescription*.

Table 5 illustrates the exemplary execution of the SPARQL query presented in Listing 1 in the Appendix on the dataset utilized in this case study. The query result demonstrates that the submitted building application represented by the returned *obpr:BuildingApplication* instance is assigned exactly one instance of the *obpr:ctContainerDescription* class. If the aforementioned query were to return an empty result set, it would be evident that no ICDD container has been assigned to the building application within the dataset under consideration.

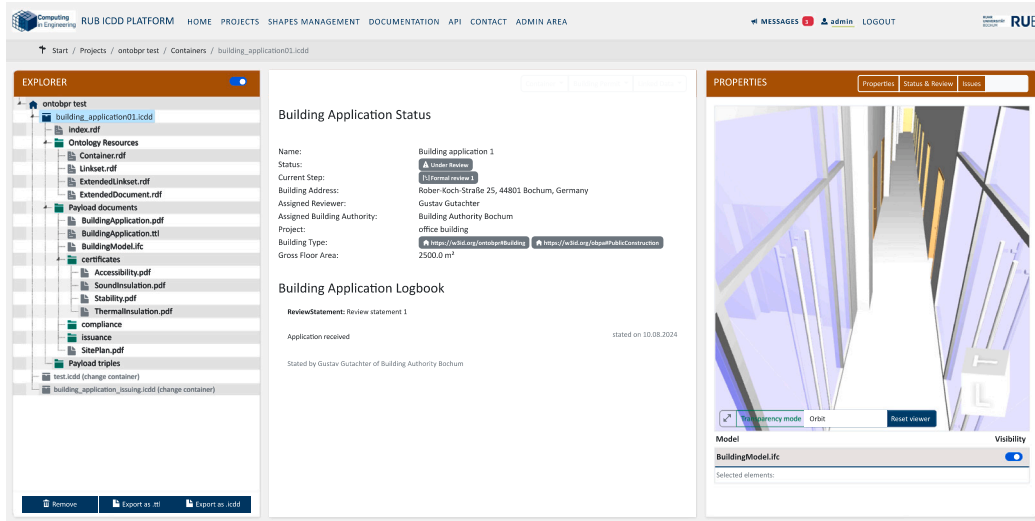


Fig. 11. Reviewing the submitted building application container in the ICDD platform.

Table 4
Competency Questions for evaluating UC1.

CQ	Question	Expected answer
CQ1.1	Is the building application delivered in an ICDD container?	ICDD Container Description
CQ1.2	Are the necessary documents for processing the application provided in the ICDD?	List of required documents
CQ1.3	Is the formal information provided in the ICDD to instantiate step 2, step 3, and step 4?	List of required building parameters

Table 5
Result of the execution of the SPARQL query for the CQ5.1 on the building application container (see Listing 1).

BuildingApp	Container
http://example.org/ontobpr-usecase#BuildingApplication_1	https://icdd.vm.rub.de/dev01/Project/RT2Xjlpuakaxob2GBRQmUg/Container/c7Rl-FbG4ECI9kww4AFuBA

Following the outcome of CQ1.1, the subsequent phase entails assessing whether the mandatory documents are present within the ICDD repository (CQ1.2). The list of the required files can be obtained from Table 2. In addition to the building model and the textual building application, this also includes, for example, certificates on the structural analyses of the building or the thermal insulation report. During the use case preparation, an object property was created for each necessary document as a sub-property of *ontobpr:hasDocument*. With the help of this set of object properties, all necessary document instances can be uniquely identified. Such sets of object properties must be created once initially for each building type. It would be conceivable to provide ICDD container templates for each building type in the future in order to simplify the submission process for both the submitter and the building authorities.

To address the competency question mentioned above, with the help of the newly created set of object properties and the developed OntoBPR framework, a SPARQL query was developed as presented in Listing 3 in Appendix. The displayed SPARQL query retrieves the information of the current building application, its related building, and its associated documents (document name and file type). The query first ensures that the selected resource is a valid instance of the class *ontobpr:BuildingApplication* and if the corresponding building instance belongs to the class *ontobpr:Building*. The following three lines ensure that the selected building instance is associated with the building application. To ensure this, the system checks whether the project linked to the building application (*obpa:includes*) with the object property

Table 6
Result of the execution of the SPARQL query for the CQ1.2 on the building application container (see Listing 3).

BuildingApp	Building	DocumentName	Filetype
:BuildingApplication_1	:office_building	BuildingApplication.pdf	pdf
:BuildingApplication_1	:office_building	BuildingModel.ifc	ifc
:BuildingApplication_1	:office_building	SitePlan.pdf	pdf
:BuildingApplication_1	:office_building	certificates/Accessibility.pdf	pdf
:BuildingApplication_1	:office_building	certificates/SoundInsulation.pdf	pdf
:BuildingApplication_1	:office_building	certificates/Stability.pdf	pdf
:BuildingApplication_1	:office_building	certificates/ThermalInsulation.pdf	pdf

obpa:hasBuilding points to the previously selected building. The query then checks for the existence of all necessary documents that are required for process step 1 (again cf. Table 2).

As presented in Table 6, the query result is the list of the necessary documents associated with the building application and its building, along with the filename and the file type for every file. This output is only generated by the query if all requested documents are available within the ICDD container. If one or more of the required documents are missing, no results are returned. This query can, therefore, be used to clearly and unambiguously check whether the ICDD container contains all the necessary documents required to process the building application provided (CQ1.2).

The competency question (CQ1.3) concerns determining whether the formal information required for instantiating steps 2, 3, and 4 is available within the ICDD container. The address, building type, and building class of the planned building are required for this instantiation. The address determines the area of responsibility of the person in charge of the building (step 2). The building and building class can be used to determine how much professional experience an expert must have (step 2), which additional external specialists must participate in the building permit (step 3), and finally, which regulations must be consulted to check the building for conformity (step 4).

Table 7

Result of the execution of the SPARQL query for the CQ1.3 on the building application container (see Listing 4).

BuildingApp	Address	BuildingClass	Type
:BuildingApplication_1	:Address01	5	ontobpr:Building
:BuildingApplication_1	:Address01	5	opba:PublicConstruction

The query formulated to obtain the address, the building type, and the building class of the corresponding building application is presented in Listing 4 in Appendix. It starts identically to the previous query for CQ1.2 by retrieving the building application under consideration and its associated building. The query also ensures that the building application is a valid instance of *ontobpr:BuildingApplication* and that the building is a valid instance of *ontobpr:Building*. It then identifies the construction project associated with the building application using the *opba:includes* property and confirms that the project is an instance of *opba:ConstructionProject*, further assuring that this project is connected to the building by the *opba:hasBuilding* property. The class of the building is retrieved through the *opba:hasBuildingClass* property, while its type is determined via the *rdf:type* property. Finally, the query retrieves the address of the building application using the *vcard:hasAddress* property. The result of the query provides the building application, its associated address, building class, and the building's type(s).

Table 7 shows the result determined with the query. Using the information obtained about the building type in steps 2, 3, and 4, it is necessary to ensure that the correct type is used for representing the building type. The results obtained with the developed query show that all formal information for the instantiating of step 2, step 3, and step 4 is available in the ICDD container, which therefore answers CQ1.3. It was shown that the competency questions could be answered using SPARQL queries and the OntoBPR framework.

7.2. (2) Assignment

Step (2) *Assignment* focuses on identifying an appropriate building official based on the location. To facilitate this, building officials need to be assigned to the application according to certain criteria. These criteria are governed by rules that reflect the specific business logic of the building authority. In this specific use case, each building official is responsible for a particular address range within the city, the building type, and the relevant subject area based on their qualifications and responsibilities. These criteria can be translated into a SHACL rule as presented in [66], which is applied to the data stored in the A-Box to determine the appropriate building official for the application. The rule ensures the building application has not yet been assigned to another building official. Furthermore, the official must have at least five years' experience if the project involves a special construction, as it is present in this case study. Finally, the building project must be within the official's area of responsibility [66].

In the given example, the authority allocates applications based on districts within the city. BO1 will assign the administrative official BO2 and technical official BO4 as the project is located in their allocated district. The allocation can be discovered through the postal code and address provided within the application form. The utilized SHACL shapes can be found in previous work [79].

7.3. (3) Participation

The step (3) *Participation* involves determining the required agencies of public interest (AoPI) that must be involved concerning their area of expertise, which is again based on the location and type of building. To facilitate this, the building permit authority must assign the relevant AoPI to the building permit under review. Because the building is near a body of water (i.e., a lake) within 500 m, an environmental expert must also be consulted. In this example, BO4 must ask the Nature

Preservation Authority Bochum for a statement (location). It is also important to note that, according to German regulations applying in this case study, public buildings and buildings with more than 1,000 m² are considered a 'special construction' and that for this type of building, the structural design documents and plans must be checked by an accredited structural engineer. For this case study, BO4 must involve an accredited structural engineer as an external expert in the audit (building type). The necessary SHACL shapes can be found in previous work [79].

7.4. (4) Content Review

Step (4) *Content Review* identifies which regulations are in place, taking into account the location and building type. Based on the location given by the postal code and address provided in the application form, the relevant regulations are applied. In this particular scenario, as explained above, it is a building in Germany, specifically in Bochum in the state of North Rhine-Westphalia (NRW). Based on this localization, the German Federal Planning Code (BauGB) [94], the NRW building code, the building code BauO NRW 2018 [77] and municipal regulations from the city of Bochum, must be applied, among others. Further, specific regulations pertinent to the building type, like the German Workplace Ordinance (Arbeitsstättenverordnung) for office buildings, are identified. Special compliance requirements, such as accessibility standards, must be adhered to for public buildings.

In this example, the building code BauO NRW 2018 [77] mandates certain requirements for the accessibility of public buildings as per §9a (Fn 9) of BauPrüfVO 1995 [93]. Regarding the layered structure of the content review conceptually defined in Fig. 5, relevant objects, such as building entrance doors, interior doors, and the outer wall concerning the plot's border, are therefore identified as part of *Layer 5* (see Fig. 5) of the review. *Layer 6* (see Fig. 5) addresses specific properties, including barrier-free access to the building, such as a ramp with a gentle slope, ensuring the minimum indoor width, and requiring that entrance doors open outwards to accommodate unrestricted wheelchair use.

This study uses particularly the example of a minimum door width requirement of at least 90 cm in public buildings that has also been presented in previous work [39], which is extended in this case study to a larger building. Detailed project information provided by the ICDD will be checked against relevant clauses, including the minimum door widths and the requirement that building entrance doors open outwards, among other specifications.

To check the minimum door width in the building model as presented in [39] and the opening directions of entrance doors, some preliminary steps are necessary to retrieve and apply the checking rules to the IFC model in the ICDD container. First, the data dictionary containing the building code for accessibility of public buildings is retrieved as *iProp:BauONRW* [77]. The building code defines a group of properties for a specific feature of interest, represented by the instance *iProp:DoorProperty*, which includes the specific instance *iProp:DoorWidth* of the type *isoprops:Property*. Verifying these properties for all doors in the building model associated with the building application is necessary. In its inherent boundary conditions, the property prescribes that doors must have a minimum width of 90 cm and that the measurements must be in centimeters.

Furthermore, an additional group of properties, represented by the instance *iProp:EntranceDoorProperty*, is required to encompass all properties that are specifically relevant to entrance doors, containing the *isoprops:Property* instance *iProp:OpeningDirection*. Parallel to the retrieval of the properties, the model provided by the applicant in IFC format is converted into LBD format within the ICDD platform. The converter used within the platform is based on the work of Oraskari et al. [36].

The property assignment of the specified properties is accomplished according to the OPM property state pattern process [38]. An example

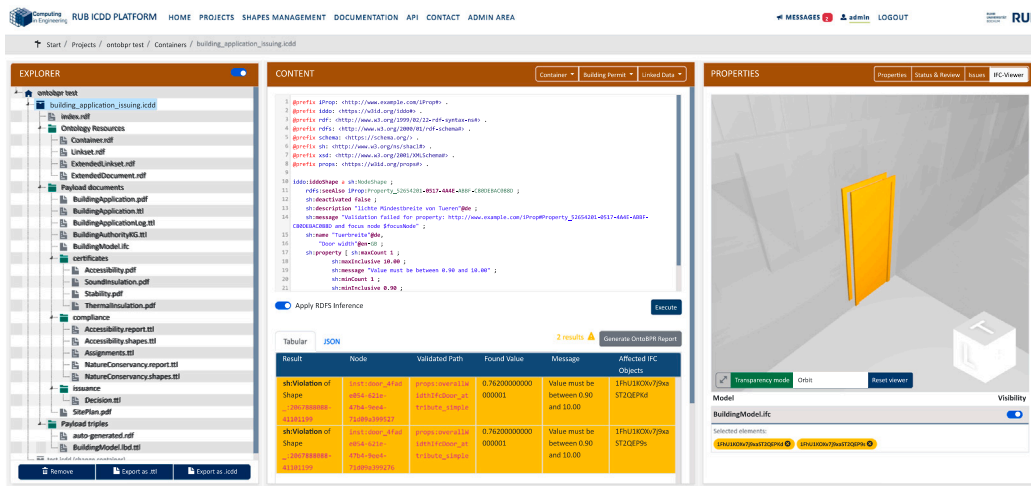


Fig. 12. SHACL validation of the building model according to accessibility.

Table 8
Competency Questions for evaluating UC5.

CQ	Question	Expected answer
CQ5.1	Are all required checks performed and their reports available in the container to draw a decision?	List of required reports
CQ5.2	If the decision is positive, are no conditions and justifications defined for the letter of notification?	List of statements with boolean literals
CQ5.3	If the decision is positive under conditions, are conditions defined for the letter of notification?	(Empty) List of conditions
CQ5.4	If the decision is negative, are justifications defined for the letter of notification?	(Empty) List of justifications

of this instruction from the public building considered in this use case can be seen in Fig. 12. Subsequently, ISOProps property constraints as SHACL shapes are created from the property sets using an ISOProps constraint generator as presented in previous work [39].

Once the SHACL shapes have been created, they are validated using the ISOProps property constraints as the shapes graph and the enriched IFC model instances as the data graph. Fig. 12 depicts an excerpt of the previously described SHACL shape (Fig. 12, top) for verifying compliance with all door widths. The results of the SHACL validation indicate that the width of two doors falls outside the specified limits (Fig. 12, below and right). The results of the SHACL validation are stored as *obpr:ReviewResult* also referring to the executing agent, while the *sh:ValidationReport* is stored inside of the container referenced in the *obpr:ReviewResult* object.

7.5. (5) Issuing Notification Letter

Step (5) *Issuing Notification Letter* involves compiling a final report based on the outcomes of the previous steps. This report will assess the completeness of the information provided, the suitability of the case officer, and the statements from other agencies. The results from the previously conducted checks are aggregated and checked for possible violations. If all interim results are positive, a positive notification will be proposed to the assigned reviewer, potentially with conditions, if any were defined by further assigned reviewers. Conversely, if any interim results are negative, a negative notification will be proposed, accompanied by justification based on the failed validations. Since the detected violations may have varying severity, the system will not automatically make decisions but only provide recommendations.

To evaluate the ontology in this step, four CQs are defined as part of the ORSD in the ontology definition step, which can be found in Table 8 with their expected answers.

To eventually draw a decision for the building application at hand, in the first step, all the results are gathered from the *obpr:ReviewResult*

Table 9

Result of the execution of the SPARQL query for the CQ5.1 on the building application container (see Listing 5 in the Appendix).

Result	Report	Conforms
:ReviewResult_1	:AcousticReport	true
:ReviewResult_2	:ThermalConductivityReport	true
:ReviewResult_4	:AccessibilityReport	false
:ReviewResult_3	:NatureConservationReport	false
:ReviewResult_5	:StructuralReport	true

class and linked to the building application via the *obpr:hasCheckingResults* property, which is also the starting point to answer the CQ5.1. The defined SPARQL query for this competency question can be found in Listing 5 in Appendix.

The query retrieves all entities that are of type *ontobpr:BuildingApplication* and have checking results. The property path *ontobpr:hasCheckingResults* is followed until reaching the *sh:ValidationReport* entity, which has attached the information about the conformance in the *sh:conforms* property. Returned values are bindings to the variables *?result*, *?report*, and *?conforms*.

As a result of the query in Listing 5, a list of reports from the application is provided in Table 9 and shows that five reports are conducted and two of them indicated that the application does not fully conform (the accessibility report and the nature conservation report). The report provides the respective severity of the violations so that a building official can make decisions on how to handle the violations, e.g., by adding conditions or rejecting the review.

Next, with the CQ5.2, added review statements and conditions or justifications can be reviewed using the query defined in Listing 6 in the Appendix. The query retrieves bindings for the variables *?statement*, *?statementType*, *?isNotCondition*, and *?isNotJustification*. With this query, it can be verified that no conditions and no justifications by reviewers are given in the container if the application is generally positively decided.

Table 10

Result of the execution of the SPARQL query for the CQ5.2 on the building application container (see Listing 6).

Statement	StatementType	isNotCondition	isNotJustification
:ReviewStatement_1	ontobpr:ReviewStatement	true	true
:ReviewStatement_4	ontobpr:ReviewStatement	true	true
:ReviewStatement_5	ontobpr:ReviewStatement	true	true
:ReviewStatement_2	ontobpr:ReviewStatement	true	true
:Condition_1	ontobpr:Condition	false	true

Table 11

Result of the execution of the SPARQL query for the CQ5.3 on the building application container (see Listing 7).

Application	Status	hasConditions
:BuildingApplication_1	ontobpr:AcceptedWithConditions	1

Therefore, as a result of the query defined in Listing 6, the list of all statement entities in the container and their respective types are provided together with a Boolean value for each of the condition or justification type assignments. We see from the results in Table 10 that four general statements are recorded in the container that are neither of type condition nor justification, and one entity is of type *ontobpr:Condition*. From these results, it can already be concluded that the planning application can only be accepted under certain conditions and can already be passed on to the decision-maker in the building authority as a proposal.

The CQ5.3 as well as CQ5.4 work in a similar way and provide the information when the decision is decided as positive with conditions (CQ5.3) or rejected (CQ5.4) that either mandatory conditions or that justifications for the rejection are defined to be able to generate the issuance letter based on the data defined in the container. The query in Listing 7 in the Appendix provides, as a result, the application, the status of the application, and the number of conditions defined for the application.

In the case of this study, the application is already considered as *ontobpr:AcceptedWithConditions* as can be seen from the previous CQs and their results. Therefore, we receive the results reported in Table 11 for CQ5.3. In Table 11, we can see that the application has one condition defined, while, in contrast, we have the empty list as a result for CQ5.4 because the filter conditions do not apply to the provided data in the building application container in this case study.

Overall, by executing the CQs on the provided dataset, we could, on the one hand, prove the applicability of the developed approach, and on the other hand, we could set the foundation for the decision-making and the issuing of the notification letter that is the substantial part of the step (5) *Issuing Notification Letter*. The queries were also stored and implemented for execution by the building officials in the web app so that they would gain the necessary insights into the building permit to conclude its status.

Based on the data, the status can be set using the implemented platform, and the decision can be defined. After the status has been set from review to a decision, the issuance letter is automatically composed as presented in Fig. 13.

In Fig. 13, the final step of issuing the notification letter in the ICDD platform is visualized. On the right-hand side, the building application status with the building application logbook containing all review statements and possible conditions or justifications. The center of the Figure contains the template for the issuance letter. In this implementation phase, the letter is put as an HTML template filled with the variables from the building application and the reviewer assignment, as well as additional information on conditions or justifications. The template can be saved as a PDF and stored inside the container so that the applicant can receive the complete container.

8. Discussion

8.1. Reflections

A key challenge we encountered during the research process was the generalizing of several different disconnected approaches that we presented in the literature in Section 2. The literature review revealed the absence of universal approaches for modeling the building permit process. Consequently, the modeling approach was initiated at a higher level of abstraction, allowing for the integration of various research approaches. However, as highlighted in previous research, such models must remain adaptable to the specific characteristics of individual systems.

The identified limitations, such as the reliance on specific regulatory formats [8,41], and the challenges with standard alignment noted by Silvennoinen et al. [42], prompted us to adopt a more flexible modeling strategy that can bridge diverse regulatory sources. While existing ontologies such as the BIM-QC Ontology of Ma et al. [45] provided valuable technical insights for code compliance checks using ontologies, their focus on technical rules rather than holistic process representation required us to provide an interface between the process ontology and already established ontologies for the compliance checking. The methodological challenges of some of the reviewed approaches stemming from inconsistent data structures, varying levels of rule formalization, and incomplete process coverage while being very well suited for covering specific tasks (for example, Fitkau et al. [11] for fire safety regulations or Zhong et al. [44] for environmental aspects) highlighted the need for a more process-aware and adaptive ontology design.

Reflecting on the methodological approach, we recognize that while the building authorities lack expertise in the Semantic Web technologies behind our developed approach, the usage of modular ontologies, an upper process ontology, and the submission of building permits in information containers can facilitate – besides other developments – the building permit review and can serve as a foundation for future developments. Modular ontologies offer a great advantage to flexibly adapt the system to country-specific legislation on the authority side, while the information containers with developed submission templates on the other side can build the bridge between the submitting party and the authorities. This experience highlighted again the need for upper ontologies and for fostering the reuse of modular ontologies, as it is described in our methodology in Section 3.

8.2. Contributions and implications

This research advances the ontology-driven digitalization of building permit reviews by introducing OntoBPR, a structured, process-oriented framework that integrates regulatory, procedural, and technical aspects of permitting into a multi-layered ontology structure. A key contribution is the formalization of a holistic model, which enables the adaptation of the framework to different legal and administrative contexts. By structuring the permit review process through horizontally aligned ontologies, OntoBPR ensures modularity and scalability, allowing for the integration of domain-specific ontologies such as those for planning law, zoning, and building regulations. The study further contributes to ontology engineering methodologies by applying the Linked Open Terms (LOT) approach to a complex regulatory process, demonstrating how ontologies can be systematically designed to represent multi-level legal and procedural structures. The multi-layer structure (Fig. 5) is particularly significant, as it provides a method to identify and formalize regulatory requirements across different levels of governance—ranging from national laws to specific building code clauses. This structured approach enhances the semantic interoperability of building permit processes.

Practitioners, including municipal authorities, regulatory agencies, and software developers, can adopt OntoBPR by integrating it into digital permitting systems, using SHACL for rule validation, and leveraging

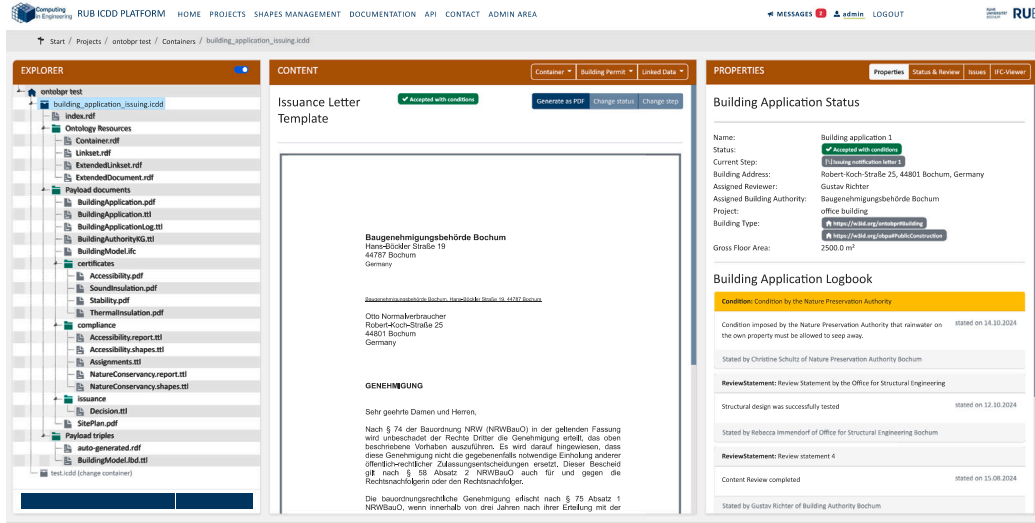


Fig. 13. Issuance of the notification letter displaying conditions for a building application that is accepted under conditions.

ICDD for data interoperability. The OntoBPR framework supports a holistic view of the permitting process, allowing fragmented steps—such as compliance checks, expert participation, and decision-making—to be seamlessly connected into a structured workflow. Additionally, OntoBPR can serve as the foundation for developing a software tool that automates compliance checking, streamlines permit processing, and enhances cross-jurisdictional regulatory alignment.

8.3. Limitations

While the OntoBPR framework presents a structured and ontology-based approach to digitalizing building permit reviews, several limitations should be acknowledged. First, despite OntoBPR being based on reliable and internationally proven datasets, it is possible that certain jurisdictions may not be fully alignable with the assumed processes and structures. Differences in administrative practices, document formats, and legal interpretations may necessitate modifications to the framework to accommodate diverse regulatory ecosystems.

Second, the framework has not yet been validated in real-life permitting processes with an actual building permit authority. Although the proposed system demonstrates conceptual feasibility and is tested with a case study, practical implementation within a municipal or regional permitting office would provide further insights into its applicability, usability, and potential challenges in real-world workflows.

Third, the integration of additional relevant ontologies into the OntoBPR framework remains an area for future work. While the current approach incorporates multiple domain-specific ontologies, expanding its scope to include additional regulatory and procedural ontologies—such as those covering environmental impact assessments, energy efficiency standards, calculating the fees, or safety regulations—could enhance its comprehensiveness.

Fourth, the case study is limited to the German regulatory environment, which, while comprehensive, does not account for variations in building codes, permit procedures, and legal frameworks across different regions or countries. Future research should explore how the framework can be adapted to jurisdictions with different regulatory structures and levels of digitalization.

Finally, an important open question concerns the development and maintenance of SHACL rules required throughout this system. While the framework proposes an automated compliance-checking process using SHACL validation, it remains unclear who within a permitting authority would be responsible for authoring, maintaining, and updating these rules. Authorities may lack the necessary expertise in ontology development and SHACL rule writing, which raises practical concerns about the

long-term feasibility and governance of such a system. Future research should investigate how permitting authorities can be equipped with the tools, training, or collaborative support needed to effectively define and manage SHACL rules, potentially involving standardized templates, domain-specific rule libraries, or Artificial Intelligence (AI)-assisted rule generation.

9. Conclusion

The primary objective of this article was to develop and validate an ontology-based framework, OntoBPR, to streamline and automate the building permit review process. This objective was accomplished through the integration of multiple domain-specific ontologies and the use of standardized information containers, specifically the ICDD to enhance the efficiency, transparency, and accuracy of building permit reviews.

This research addressed three key questions. First, it explored how digital building applications could be submitted and reviewed in a unified framework (as defined by RQ1). The OntoBPR framework facilitates this concept by utilizing ICDD containers and multiple layers of ontologies, which structure the information consistently. This enables seamless data exchange and automated compliance checking through SHACL rules and ensures that all necessary information is included and correctly formatted, significantly reducing the manual effort required for review. Second, the framework addressed how the completeness of digital building applications could be verified and prepared for further processing within building authorities (as defined by RQ2). By embedding SHACL validation rules within the ICDD containers, the framework automatically verifies the presence and correct structure of all required information during the formal review process, preparing the application for subsequent steps without unnecessary delays or errors. Lastly, the paper examined how results of digital building permit reviews could be gathered to support building officials' decision-making processes and how digital notification letters could be automatically generated using OntoBPR (as defined by RQ3). Therefore, the framework collects validation reports within the ICDD containers, and consequently supports automated reasoning to facilitate decision-making, however, it is important to state, that no decisions are drawn by the system but only recommendations given based on the number and severity of violations registered in the report. The system enables applicants to receive timely and accurate updates on their permit applications by generating semi-automated notification letters.

The framework was validated through a detailed case study demonstrating its capability to handle a typical building permit application

using ICDD containers. The validation process involved utilizing SHACL rules within the ICDD container to automate compliance checking and assess the completeness of the provided information. The results showed that the OntoBPR framework, supported by the ICDD container approach, could effectively process building permit applications, ensure compliance with relevant regulations, and automatically generate necessary documentation. Compared to traditional permit review processes, which are often highly manual and prone to inconsistencies, OntoBPR provides a systematic and automated framework that reduces manual effort, ensures completeness of applications through standardized validation, and enhances transparency. Unlike existing ontology-based methods, OntoBPR uniquely integrates multiple domain ontologies with ICDD containers, ensuring end-to-end consistency and adaptability to various regulatory environments. Even though there is still much research to be done on the digital transformation of building permitting, OntoBPR offers a novel approach and a first step to integrating the fragmented elements of the building permit process.

While the OntoBPR framework represents a significant advancement in automating building permit reviews, several areas require further exploration. Future research must focus on expanding the scope of the ontologies used to cover additional regulatory aspects and more diverse building types, thereby improving the framework's applicability across different regions and project scales. Implementing pilot programs with real-world building authorities would provide valuable insights into user adoption, challenges, and areas for improvement, with feedback from practitioners essential for refining the framework. Finally, adapting OntoBPR for use in different legal and regulatory environments, including cross-jurisdictional permit reviews, would further demonstrate its flexibility and utility. Utilizing the ICDD, the log stored within the ICDD container could be used in future research to conduct process mining analyses. By addressing these areas in future research, OntoBPR can be further refined and expanded, paving the way for more widespread adoption and contributing to the ongoing digital transformation of building permits.

CRedit authorship contribution statement

Philipp Hagedorn: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Judith Fauth:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Sven Zentgraf:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sebastian Reiß:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Markus König:** Writing – review & editing, Writing – original draft, Supervision, Project administration. **Ioannis Brilakis:** Writing – review & editing, Writing – original draft, Supervision, Project administration.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Grammarly AI writing assistance in order to facilitate spell and grammar checking. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A.1. Requirements

See Table 11.

A.2. Queries

```
1 SELECT ?buildingApp ?container
2 WHERE
3 {
4   ?buildingApp a ontobpr:BuildingApplication .
5   ?buildingApp ontobpr:hasBuildingApplication- Container ?container .
6   ?container a ct:ContainerDescription .
7 }
8 HAVING (COUNT(?container) = 1)
```

Listing 1: SPARQL query for competency question CQ1.1 .

```
1 obpr:BuildingApplicationShape
2 a sh:NodeShape ;
3 sh:targetClass obpr:BuildingApplication ;
4 sh:property [
5   sh:path obpr:hasBuildingApplication- Container ;
6   sh:minCount 1 ;
7   sh:maxCount 1 ;
8   sh:class ct:ContainerDescription; ] .
```

Listing 2: SHACL rule for competency question CQ1.1 .

```
1 SELECT ?buildingApp ?building ?documentName ?filetype
2 WHERE {
3   BIND(ontobpr-usecase:BuildingApplication_1 AS ?buildingApp) .
4   BIND(ontobpr-usecase:office_building AS ?building) .
5
6   # Check whether the resources belong to the right class
7   ?buildingApp a ontobpr:BuildingApplication .
8   ?building a ontobpr:Building .
9
10  #Check whether the application and the building are linked via the project
11  ?buildingApp obpa:includes ?project .
12  ?project a obpa:ConstructionProject .
13  ?project obpa:hasBuilding ?building .
14
15  # Ensure that all documents are available
16  ?buildingApp ontobpr:hasApplicationDocument ?applicationDocument .
17  ?buildingApp ontobpr:hasSitePlanDocument ?sitePlanDocument .
18  ?buildingApp ontobpr:hasStabilityDocument ?stabilityDocument .
19  ?buildingApp ontobpr:hasSoundInsulationDocument ?soundInsulationDocument .
20  ?buildingApp ontobpr:hasThermalInsulationDocument ?thermalInsulationDocument .
21
22  ?buildingApp ontobpr:hasAccessibilityDocument ?accessibilityDocument .
23  ?building ontobpr:hasBuildingDocument ?buildingDocument .
24  # Output of document names and file types for each document
25  {
26    ?applicationDocument ct:name ?documentName ;
27    ct:filetype ?filetype .
28  }
29  UNION
30  {
31    ?sitePlanDocument ct:name ?documentName ;
32    ct:filetype ?filetype .
33  }
34  BIND("Site Plan Document" AS ?documentType)
35  ...
36  ORDER BY ?documentName
```

Listing 3: SPARQL query for competency question CQ1.2 .

Table 12
Ontology requirements specification document (template provided by [95]).

Ontology Requirements Specification Document	
1. Purpose	
OntoBPR supports the building permit review process by using and aligning existing ontologies for a transparent, efficient, and objective administrative procedure, ensuring legal compliance. OntoBPR aims to support the building official with semi-automated checks.	
2. Scope	
OntoBPR includes the quality-assured and seamless exchange of information between the different steps of the building permit review.	
3. Implementation Language	
Web Ontology Language (OWL)	
4. Intended End-Users	
Intended end users are persons working in the processes: building authorities processing the building permits and specialist inspectors involved through the participation process. Finally, the building applicants should also be considered as end users.	
5. Intended Use Cases (UC)	
1. Formal Review UC1 Assignment (already implemented) UC2 Participation (already implemented) UC3 Content Review (already implemented) UC4 Issuing Notification Letter	
6. Ontology Requirements	
a. Non-Functional Requirements (NFR)	
NFR1 Coverage and Sufficiency: This pertains to the breadth and depth of the ontology's domain representation. NFR2 Consistency: This ensures the ontology remains free from contradictions and conflicts. NFR3 Usability: This aspect focuses on the ease of use and practicality of the ontology in relevant applications. NFR4 Clarity and Conciseness: This NFR emphasizes the ontology's understandability and succinctness in its representation. NFR5 Reusability and Extensibility: This refers to the ontology's ability to be adapted for various purposes and its capacity for expansion. NFR6 Transparency: This criterion underlines the need for the ontology to be understandable and transparent in its structure and logic. NFR7 Modularity: This focuses on the ontology's design in discrete, manageable sections for ease of understanding and maintenance.	
b. Functional Requirements:	
Lists of requirements written as competency questions (CQs) and sentences	
UC1: Formal Review	UC5: Issuing Notification Letter
CQ1.1 Is the building application delivered in an ICDD container? CQ1.2 Are the necessary documents for processing the application provided in the ICDD? CQ1.3 Is the formal information provided in the ICDD to instantiate Step 2, Step 3, and Step 4?	CQ5.1 Are all required checks performed and their reports available in the container to draw a decision? CQ5.2 If the decision is positive, are no conditions and justifications defined for the letter of notification? CQ5.3 If the decision is positive under conditions, are conditions defined for the letter of notification? CQ5.4 If the decision is negative, are justifications defined for the letter of notification?

```

1 SELECT ?buildingApp ?address ?buildingClass ?type
2 WHERE {
3
4   BIND(ontobpr-usecase:BuildingApplication_1 AS ?buildingApp) .
5   BIND(ontobpr-usecase:office_building AS ?building) .
6
7   # Check whether the resources belong to the right class
8   ?buildingApp a ontobpr:BuildingApplication .
9   ?building a ontobpr:Building .
10
11  #Check whether the application and the building are linked via the project
12  ?buildingApp obpa:includes ?project .
13  ?project a obpa:ConstructionProject .
14  ?project obpa:hasBuilding ?building .
15
16  ?building obpa:hasBuildingClass ?buildingClass .
17  ?building a ?type .
18  ?buildingApp vcard:hasAddress ?address .
19 }
20 ORDER BY ?documentName

```

Listing 4: SPARQL query for competency question CQ1.3 .

```

1 SELECT ?result ?report ?conforms
2 WHERE {
3   ?application a ontobpr:BuildingApplication.
4   ?application ontobpr:hasCheckingResults ?result.
5   ?result ontobpr:hasCheckingResults ?report.
6   ?report sh:conforms ?conforms.
7 }

```

Listing 5: SPARQL query for competency question CQ5.1

```

1 SELECT ?statement ?statementType ?isNotCondition ?isNotJustification
2 WHERE {
3   ?subject a ontobpr:BuildingApplication.
4   ?subject ontobpr:hasStatement ?statement.
5   ?statement a ?statementType.
6   BIND(?statementType!=ontobpr:Condition as ?isNotCondition)
7   BIND(?statementType!=ontobpr:Justification as ?isNotJustification)
8   OPTIONAL{
9     ?subject ontobpr:hasStatus ontobpr:Accepted.
10  }
11 }

```

Listing 6: SPARQL query for competency question CQ5.2

```

1 SELECT ?application ?status (COUNT(?statement) as ?hasConditions)
2 WHERE {
3   ?application a ontobpr:BuildingApplication.
4   ?application ontobpr:hasStatus ?status.
5   ?application ontobpr:hasStatement ?statement.
6   ?statement a ?statementType.
7   FILTER(?status=ontobpr:AcceptedWithConditions && ?statementType=
   ontobpr:Condition).
8   #FILTER(?status=ontobpr:Rejected && ?statementType=ontobpr:Justification).
9 } GROUP BY ?application ?status

```

Listing 7: SPARQL query for competency question CQ5.3 and CQ5.4

Data availability

The data produced and used in this article is made available through a public repository at: <https://github.com/RUB-Informatik-im-Bauwesen/ontobpr>.

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